

Refining *find*

Natasha Korotkova (Utrecht University) and Pranav Anand (UCSC)

August 2026

Abstract This paper focuses on a series of puzzles in the semantics of the so-called subjective attitudes, zooming in on the behavior of English *find* combined with small clause complements. *Find*-verbs, as we will call them, have been important in understanding subjectivity in natural language: they select for subjective complements (*Q finds humans ✓entertaining/#mortal*) and their very existence was often used as an argument in favor of treating subjective predicates (e.g., *entertaining* but not *mortal*) as having a special semantics. To this end, *find*-verbs have been largely regarded as subjective counterparts of *think*-verbs. Focusing on English *find*, we challenge this view and show that *find* exhibits a distinguished linguistic profile not shared by *think*, including subject-predicate asymmetries in the embedded clause, the lack of referential opacity and the similarity to emotive factives, aspectual class and the acquaintance requirement. We propose a principled account of these seemingly unrelated properties by appealing to the small clause syntax and the lexical semantics of *find*, including its argument structure and the nature of eventuality it describes. We argue that some idiosyncratic features of *find* follow naturally once it's seen within a broader context of what we dub 'characterizing verbs', which express evaluation and take structured proposition arguments. We further show that, based on the interpretation of referential DPs in its scope, *find* is best understood as a knowledge verb, not as a belief verb. Finally, we establish that several properties of *find* are due to the fact that it is an achievement predicate describing a complex event of judgment formation caused by an event of encounter. We remain deliberately agnostic about the exact nature of the attitude *find* encodes: our account is compatible both with views under which *find* encodes a belief and with views under which *find* encodes a non-cognitive, affective state. By treating *find* first and foremost as a verb, and not only as a litmus test for subjectivity, we aim to emphasize the importance of linguistic data for the debates in philosophy of language and to contribute to the burgeoning research agenda on the fine-grained semantics of attitude predicates.

Keywords achievement, acquaintance inference, attitude predicates, *find*, knowledge ascriptions, lexical semantics, subjective language

1	Introduction	3
2	Theoretical background	5
2.1	Subjective language	5
2.2	<i>Find</i>	6
2.3	<i>Find</i> vs. <i>consider</i>	7
2.4	Scope of the paper	9
3	Core data	10
3.1	Subject-predicate asymmetries and structured meaning	10
3.2	Referential transparency	15
3.3	Factivity	19
3.4	Aspectual class	22
3.5	Acquaintance requirement	25
3.6	Recap	30
4	Analysis	30
4.1	Structured propositions and Sæbø's Puzzle	32
4.2	The semantics of <i>find</i> : aspect and experience	33
4.3	The stative <i>find</i> conspiracy	36
4.4	Non-subjective content: making the right cut	39
5	Loose ends	42
5.1	Causality	42
5.2	Vicarious experience and vividness	45
6	Outlook	46

1 Introduction

This paper is a focused investigation of the English verb *find* when it combines with small clause complements. *Find*-verbs in English and several other European languages have featured prominently in the literature on subjective language and have often been called ‘subjective attitudes’: they allow subjective complements (which talk about matters of opinion), but not objective complements (which talk about matters of fact). With some exceptions (Franzén 2020, 2021; Fusco et al. 2025; Khoo 2026; Sæbø 2009), much of the literature treats *find*-verbs as a species of *think*-verbs: doxastic attitude predicates that pose additional restrictions on their complements (Bouchard 2012; Coppock 2018; Kennedy & Willer 2016, 2022; Korotkova & Anand 2021; Muñoz 2019; Stephenson 2007b). (1) illustrates this key contrast for English: while *think* is compatible with subjective and objective complements alike (1a), *find* requires a subjective complement that can be construed as a matter of opinion (1b).

- (1) *Q is an omnipotent being with infinite lifespan within Star Trek Universe.*
- a. *Q thinks that humans are ✓entertaining/✓mortal.*
 - b. *Q finds humans ✓entertaining/#mortal.*

Our goal in this paper is to shift the focus of attention away from subjectivity (about which we will say little) to a series of empirical puzzles about English *find* that so far have received far less focus in the existing literature. Throughout the paper, we will be concerned exclusively with the subjective *find* (1b) and will not discuss *find*’s other lives, e.g., its uses with DP complements (*to find happiness*) or with full clauses (*to find that Vulcan has three suns*), neither of which require the complement to be subjective.^{1,2} We focus on the following properties of our *find*:

- A: subject-predicate asymmetries: *find*, but not *think*, requires the subjective content to be part of the predicate position of the embedded clause (first pointed out by Sæbø 2009);
- B: referential transparency and factivity: *find*, but not *think*, blocks a true ‘de dicto’ reading for non-subjective content in its complement and behaves similarly to emotive factives (some, but not all, of these data are discussed by Bouchard 2012; Coppock 2018; Khoo 2026);
- C: lexical aspect: *find*, but not *think*, is an achievement—a punctual change-of-state predicate (pace Vardomskaya 2018);
- D: the acquaintance requirement: *find*, but not *think*, requires its subject to have firsthand experience with the entity in question (first discussed by Borkin 1973; Ducrot 1975; Stephenson

1. The literature on subjectivity with English *find* and *consider* has focused almost exclusively on small clause complements. The lone exception is Bouchard (2012), who argues that *find* is subjective even with declarative *that*-clauses, as in *Q finds that humans are entertaining*. Nevertheless, Bouchard acknowledges that *find* + *that*-clauses has no subjectivity requirement when “used to describe a discovery” (Bouchard 2012:185, fn.18), presumably suggesting that there are two instances of *find that*, one subjective and one non-subjective. However, the distinction between these two uses amounts principally to whether the complement is subjective, and so it becomes hard to separate them in a non-circular way. In the rest of the paper we will confine our attention only to small clause complements of *find*, which we maintain must contain a subjective expressions. We also put aside special uses of *find* + small clauses in the context of legal proceedings, such as *The jury found him innocent*.

2. There is an important strand of research that we do not discuss here. It concerns the exact nature of the interaction between clause-taking predicates and the syntactic shape of their complements (Bogal-Albritten 2016; Moulton 2009; Özyıldız 2017; Wurmbrand 2024; Wurmbrand & Lohninger 2023), including complement-driven meaning alternations such that a predicate can have different interpretations—and possibly denotations—based on the syntactic type of its complement (Barwise 1981; Higginbotham 1983; Moulton 2009). We are open to the possibility that different readings of *find* are influenced by different types of complements it can combine with, but don’t explore this idea here.

2007b and argued for in detail by Korotkova & Anand 2021).

We propose a principled account of these seemingly unrelated properties by appealing to the small clause syntax and the lexical semantics of *find*, including its argument structure and the nature of the eventuality it describes.

First, we start with an observation that *find* belongs to a class of what we dub ‘characterizing verbs’ (including *consider*, but also *deem*, *judge*, *regard*), predicates which (i) syntactically embed a small clause, and (ii) semantically present an entity (the small clause subject) and express an evaluation of that entity (the small clause predicate). While characterizing verbs differ in the nature of that evaluation, they all give rise to subject-predicate asymmetries in the complement clause that was first pointed out by Sæbø (2009) for *find*. Inspired by prior work on these verbs within syntactic literature (Basilico 2003), we propose that they all require their small clause complement to be a structured proposition (in line with Bittner 1998a,b on *consider*), which allows us to solve Sæbø’s (2009) puzzle without running into the compositional problems of his account. Furthermore, capitalizing on the vast body of work that connects interpretational constraints on noun phrases and information structure (Cohen & Erteschik-Shir 2002; Jäger 2001; Kuroda 1972; Ladusaw 1994), we suggest that all characterizing small clauses have a topic-comment structure.

Second, we show that *find* does not exhibit standard intensional effects with respect to the interpretation of referential DPs in its scope, neither is it easily amenable to a purely extensional treatment. The complex pattern is that subjects of the embedded clause are always interpreted ‘de re’ while the predicate position is split into objective content (that must be accurate both at the world of evaluation and in the finder’s belief worlds) and subjective content (that need not hold at the world of evaluation). We argue that this behavior closely matches that of emotive factives (*be amused*, *glad*, *sad*, *surprised*) and that *find* is a knowledge verb, not a belief verb.

Finally, we argue that as an achievement, a *find* event picks out a change of state, wherein an event of encounter causes the formation of a novel judgment.³ This analysis links the lexical aspect of *find* with its acquaintance requirement in a principled way. Importantly, the acquaintance requirement should not be construed simply as an evidential add-on to a doxastic judgment (as in Korotkova & Anand 2021; Stephenson 2007b). Instead, it should be understood as a reflection of the event structure, which encodes the encounter as a causal grounding for the judgment.

Overall, by treating *find* first and foremost as a verb, and not only as a litmus test for subjectivity, we aim to emphasize the importance of linguistic data for the debates in philosophy of language and to contribute to the burgeoning research agenda on the fine-grained semantics of attitude predicates.

We proceed as follows. §2 provides a primer on subjective language, outlining the relevant aspects of existing proposals for *find*. §3 is the empirical foundation of the paper. It goes over *find*’s special properties that distinguish it from *think* and that, in their entirety, are neither discussed nor predicted by current accounts: small clause complements (§3.1), referential transparency (§3.2), factivity (§3.3), lexical aspect (§3.4), and the acquaintance requirement (§3.5). §4 lays out a novel analysis of *find* that treats it as an intensional, but not necessarily a doxastic predicate. Our goal is to provide a principled explanation of the distinctive behavior of *find* that so far escaped systematic attention. §5 outlines several remaining puzzles, along with pointers to possible solutions. §6 concludes with an outlook of how *find* fits in the var-

3. In §4.3, we show that *find* may look stative (and thus more similar to *think* than it actually is) because many previous examples are in the simple present like (1b), which allows a habitual reading of *find* under instances of multiple encounters.

ied landscape of attitude ascriptions and with a discussion of how *find*-verbs can be identified across languages, using English *find*'s many special properties as a starting point.

2 Theoretical background

2.1. Subjective language

There is an intuitive difference between ‘subjective concepts’, which describe matters of opinion,⁴ and ‘objective concepts’, which describe matters of fact. Subjective concepts are intricately linked to subjective language, and the linguistic behavior of predicates expressing opinion has often been used to motivate different views on subjectivity in language and thought. For the purposes of this paper, this varied theoretical landscape can be divided into two families: exceptionalism and unexceptionalism (see recent overviews in [Anand & Korotkova 2022](#); [van Wijnbergen-Huitink 2016](#)).⁵

The exceptionalist argues that predicates expressing subjective concepts, e.g., *delicious*, have special denotations that distinguish them from ordinary predicates: e.g., in addition to being sensitive to the worlds of evaluation such predicates also make reference to a judge/assessor in their semantics or have an otherwise different semantic type ([Bylinina 2017](#); [Coppock 2018](#); [Egan 2010](#); [Lasersohn 2005](#); [MacFarlane 2014](#); [Pearson 2013a](#); [Sæbø 2009](#); [Stephenson 2007b](#); [Stojanovic 2007](#)). This view is often motivated by the data on faultless disagreement ([Kölbel 2004](#); [Zakkou 2019](#)), a phenomenon whereby the interlocutors can have opposing beliefs about the truth value of a given proposition without either of them being wrong. (2) is a dispute about facts: for any given tea, it can be from Yunnan or from elsewhere, but not both at the same time, so only A or only B can be right, but not both at the same time. (3) is different: what counts as delicious is in the eye of beholder, and both A and B can be right at the same time without a contradiction, hence the term ‘faultless’. Such contrasts are taken to indicate that (i) *delicious*, but not *being from Yunnan*, is a subjective predicate and that (ii) personal taste, but not geographical origin, is a subjective concept.⁶

- (2) A. This puerh tea is from the Yunnan province of China. [Ordinary disagreement]
 B. No, it's isn't.
- (3) A. This puerh tea is delicious. [Faultless disagreement]
 B. No, it isn't.

However, faultless disagreement may not be a good tool to diagnose subjectivity, as it arises not only with intuitively subjective predicates like *delicious* (3), but with a wide range of expressions that deal with vagueness and contextual underspecification ([Barker 2013](#); [Glanzberg 2007](#); [Kennedy 2013](#); [Solt 2018](#); [Plunkett & Sundell 2013](#)). The unexceptionalist takes such data seriously and argues that predicates like *delicious* are not special semantically ([Kennedy & Willer 2016, 2022](#)), or maybe not special at all and can be assimilated to ordinary intersective

4. The literature often uses the term ‘evaluative’ interchangeably with, or instead of, ‘subjective’.

5. We deliberately ignore conceptual differences between contextualism, relativism and expressivism, as they are irrelevant here; see [MacFarlane \(2014\)](#) for discussion.

6. Many subjective predicates can take prepositional phrases that overtly express the assessor, as in *delicious to me*. Such uses do not give rise to faultless disagreement and are not embeddable under *find*, which is often accounted for by treating such prepositional phrases as overtly expressing an implicit argument present in the semantics of *delicious* ([Bylinina 2017](#); [Collins 2013](#); [Stephenson 2007a](#)). We will not discuss such uses further and simply note that there are asymmetries between *delicious* vs. *delicious to me* that are difficult to explain if they have the same underlying semantics ([Anand & Korotkova 2018, 2022](#)).

predicates modulo different types of context-sensitivity.

2.2. Find

Find-verbs have featured prominently in the debates on subjective language. Often referred to as ‘subjective attitudes’ (term due to [Kennedy & Willer 2016](#)), their very existence has been used as a strong argument in favor of some form of exceptionalism. Many European languages have verbs that, like English *find* (1b), select for subjective complements: Catalan *trobare* and Dutch *vinden* ([Korotkova & Anand 2021](#)), French *trouver* ([Bouchard 2012](#); [Ducrot 1975](#)), German *finden* ([Frühauf 2015](#); [Lande 2009](#); [Reis 2013](#)), Norwegian *synes* ([Lande 2009](#); [Sæbø 2009](#)), Portuguese *achar* ([Mendes & Kendrick 2024](#)), or Swedish *tycka* ([Coppock 2018](#)). Norwegian and Swedish are particularly interesting. Unlike English *find*, which yields different meanings based on the type of complement it combines with, *synes* and *tycka* always take full clauses and seem to be earmarked for expressing one’s opinion. (4) illustrates.

- (4) Magda synes at [Norwegian]
 Magda be.of.opinion that
 kjempesequoiaetre-et er et ✓elegant/#evergreen tre.
 giant.sequoia.tree-the be a elegant/evergreen tree
 ≈‘Magda finds the giant sequoia an ✓elegant/#evergreen tree.’
 (adapted from [Korotkova & Anand 2021:516](#))

Inquiry into *find* and its counterparts in other languages has primarily focused on ferreting out what *find*-verbs reveal about subjectivity and how the grammar conceptualizes the subjective-objective divide.⁷

According to the line of thought pioneered by [Stephenson \(2007b\)](#), *find* is a subjective cousin of *think*: a doxastic attitude predicate that introduces an intensional environment and supplements the semantics of *think* with a selectional requirement that its propositional complement is subjective (and other constraints; §3.5).

- (5) $\llbracket \text{find} \rrbracket^w = \lambda x \lambda p : p \text{ is subjective. } \forall w' \in \text{DOX}(x, w)[p(w')]$.

In the spirit of [Lasersohn \(2005\)](#) and [MacFarlane \(2014\)](#), this view typically assumes a type distinction between subjective vs. objective predicates ([Bouchard 2012](#); [Bylinina 2017](#); [Coppock 2018](#); [Stephenson 2007b](#)), though it is compatible with different construals of subjectivity as long as there is something—in semantics or pragmatics—that differentiates *elegant* from *evergreen* ([Kennedy & Willer 2016, 2022](#)).

7. Embedding under *find* is often seen as a reliable diagnostic of subjectivity, especially in contrast to faultless disagreement (see discussion in [Anand & Korotkova 2022](#); [Kennedy 2013](#)). While we will not address this issue in detail here, we want to point out that it is a one-way diagnostic: it is not a given that all predicates that are banned under *find* are not subjective. Consider the following. On the one hand, subjective predicates in English are often argued to be individual-level ([Pearson 2013a](#)), a type of predicate that expresses essential properties and is characterized by special linguistic behavior such as the incompatibility with temporal modifiers ([Carlson 1980](#) and later work, though see [Jäger 2001](#) for a more fine-grained picture). On the other hand, subjectivity is often determined based on a predicate’s ability to appear under *find*. This leaves the following gap: are there subjective predicates that are non-individual-level and cannot appear under *find* because *find* selects for individual-level predication? ([Pearson \(2022\)](#) discusses *tastes good* as an example of non-individual-level subjective predicates, however, such expressions are bad under *find* for syntactic reasons—they are verb-based.) Here is a range of predicates that could be construed as subjective based on the concepts they express, but that are nonetheless infelicitous under *find*: *fetid*, *languid*, *putrid*, *tepid*, *torpid*. We come back to this issue in §3.1; see also [Korotkova & Anand 2021](#) for other constraints on embedding under *find*.

According to another line of thought, *find* is a non-doxastic extensional predicate that expresses an affective attitude (Franzén 2020, 2021; Khoo 2026), in line with expressivist theories that treat subjective language at large on a par with expressive speech acts such as exclamations (Gibbard 1990). For our purposes, the key insight of this view is that *find* is radically different from *think* because an attitude expressed by *find* is not a belief at its core, though it may result in a belief. While we are sympathetic to the idea that judgment is ontologically distinct from belief and that there are linguistic expressions that are sensitive to this distinction (for example, the class of characterizing verbs we discuss in §3.1), in this paper we will focus on the linguistic behavior of *find* as a necessary step towards understanding the concepts its lexicalizes; see Khoo (2025) for a detailed discussion of judgment.

Finally, the reductionist account in Sæbø (2009) dispenses with the idea that *find* lexicalizes an attitude at all. According to Sæbø, all *find* does is explicitly fix the individual whose opinion is conveyed by subjective expressions in its scope. It takes a property argument, which is itself a two-place predicate with one position filled by the embedded subject, and an individual argument, which fills the second position of the embedded predicate (6b). According to Sæbø, subjective predicates are judge-dependent (they have special denotations relative to an additional parameter of evaluation) and *find* simply switches the order of composition of the arguments for subjective expressions. As a result, (6a) and (6c) have the same denotation in (6d).

- (6) a. Q finds (Catherine) Janeway entertaining.
 b. $\llbracket \text{entertaining} \rrbracket^w = \lambda x \lambda j \lambda s. \text{entertaining}'(x, j, s, w)$
 $\llbracket \text{find} \rrbracket^w = \lambda P_{\text{sect}} \lambda x. P(w)(x)$
 $\llbracket \text{find Janeway entertaining} \rrbracket^w = \llbracket \text{find} \rrbracket^w (\lambda w. \llbracket \text{Janeway entertaining} \rrbracket^w)$
 $= \llbracket \text{find} \rrbracket^w (\lambda w \lambda j \lambda s. \text{entertaining}'(\mathbf{J}, j, s, w))$
 $= \lambda x \lambda s. \text{entertaining}'(\mathbf{J}, x, s, w).$
 c. Catherine Janeway is entertaining to Q.
 d. $\llbracket (6a) \rrbracket^w = \llbracket (6c) \rrbracket^w = \lambda s. \text{entertaining}'(\mathbf{J}, \mathbf{Q}, s, w)$

2.3. Find vs. consider

One common thread in this literature is to juxtapose *find*, *think* and an additional verb, *consider*, on the scale of subjectivity: *find* and *consider* are construed as special versions of *think*, with *find* a more restricted variant of *consider* (Coppock 2018; Kennedy & Willer 2016, 2022). The paradigm in (7)–(9), adapted from (Kennedy & Willer 2016:914), illustrates. Both *find* and *consider* allow intuitively subjective complements, such as *fascinating* (7a,8a), and block intuitively objective complements, such as *equal to four* (7c,8c). *Consider* (8b), but not *find* (7b), allows complements that can be construed as having more than one meaning contextually, such as *vegetarian*. Vanilla doxastics *think* and *believe* are omnivorous and as such felicitous in all three cases (9).

- (7) Kim finds ...
 a. ✓...Lee fascinating.
 b. #...Lee vegetarian, because the only animals he eats are oysters.
 c. #...the sum of two and two equal to four.

- (8) Kim considers ...
- a. ✓...Lee fascinating.
 - b. ✓...Lee vegetarian, because the only animals he eats are oysters.
 - c. #...the sum of two and two equal to four.
- (9) Kim thinks/believes that ...
- a. ✓...Lee is fascinating.
 - b. ✓...Lee is vegetarian, because the only animals he eats are oysters.
 - c. ✓...the sum of two and two is equal to four.

Based on the contrasts in (7)–(9), [Kennedy & Willer \(2016, 2022\)](#) argue that *find* and *consider*, but not *think*, presuppose the counterstance-contingency of their complements. Counterstance-contingency is a species of contextual indeterminacy: for any given information state that resolves factual issues in the same way, there are alternative information states that differ in the resolution of vague or otherwise contextually unsettled issues. For example, an agent’s information state can be decided on whether Lee eats oysters and its alternative states—counterstances—only differ in whether or not oyster-eating is compatible with being vegetarian (e.g., it is not if ‘vegetarian’ only encompasses those individuals who consume no animals). Expressions whose interpretation is sensitive to non-factual resolution are counterstance-contingent, and *consider* is argued to require precisely such complements. A related approach along similar lines is due to [Muñoz \(2019\)](#), who treats *consider*-complements as hyperintensional. Additionally, the more restrictive *find* requires radical counterstance-contingency: the issues raised by *find*-complements cannot be resolved via ordinary conversational means (such as agreeing what ‘vegetarian’ amounts to) and thus constitute genuine matters of opinion that are irreducible to contextual indeterminacy or metalinguistic disagreement. We believe that this is a different implementation of the core intuition that *find* requires subjective complements. The key difference from many previous approaches is as follows: subjectivity (or radical counterstance-contingency) is viewed not as a property of lexical items that is baked into their conventional meaning, but as a property of an agent’s information state that may depend on contextual factors in addition to lexical meaning.

Our primary concern is *find*, so we will say little about *consider*. However, we want to point out the following prediction of Kennedy & Willer’s account (modulo the formal differences, the accounts in [Coppock 2018](#) and [Muñoz 2019](#) are similar). It is clear that *consider* is sensitive to how different lexical expressions can be construed in a given context and not just to their lexical semantics. *Find* is assumed to be similar, since *prima facie* *find*-complements constitute a proper subset of *consider*-complements. The conclusion is that the semantics of *find* is underpinned by a more restricted version of the same semantic mechanism as the one for *consider*. Because *consider* requires counterstance-contingency but not the absence of radical counterstance-contingency, this view predicts that *consider* will be synonymous with *find* with radically counterstance-contingent complements settled via personal experience and not conversational negotiation. In other words, (8a) is predicted to have a reading equivalent to (7a). As discussed by [Soria-Ruiz \(2021\)](#), this prediction is incorrect. When *consider* is used with canonical evaluative adjectives, including taste predicates like *delicious*, it is about matters of metalinguistic dispute and not about personal opinion (cf. also [Stojanovic & McNally 2023](#) on the interpretational differences of moral predicates under *find* vs. *consider*). Kennedy and Willer correctly predict that *consider* will have this reading; the problem is that it is also predicted to have another, unattested reading such that *consider* is equivalent to *find*.⁸ We treat

8. [Kennedy & Willer \(2022:1436-1439\)](#) assume that the interpretational differences between *find* and *consider*

such data as casting doubt on the idea that *find* is a more restricted version of *consider* with the same basic semantics; see also [Fusco \(2025\)](#) for a similar point about *consider* vs. *think*.

2.4. Scope of the paper

Summing up, the discussion of *find* has proceeded along the following axes: what kinds of predicates does *find* embed? what is the relationship between *find* and *consider*? are both of them just special versions of *think*, with *find* being a more restricted species of *consider*? is *find* intensional or extensional? does it express an attitude and if it does, is it doxastic? While we will take a stand on some of those issues, our overarching goal is to offer a novel theory of *find* that is based on a range of properties that have received less attention in recent literature. To this end, there are issues that we will not be discussing, or only discussing in passing, as they are not central to our core proposal. In other words: we are content to leave some puzzles about *find* not fully resolved for now, in hopes that they will be addressed in greater detail in future research. Below we delineate several issues we will not be concerned with.

First, subjectivity, which has been front and center in existing literature, will not concern us here. The choice between current theories will be inconsequential for most of the paper (though see [Anand & Korotkova \(2022\)](#) for compositional constraints on theories of subjective meaning).

Second, we would like to disentangle intensionality from the nature of the attitude. Current literature offers the following options: *find* is treated as either (i) intensional & doxastic, or (ii) extensional & non-doxastic/affective. However, these properties are conceptually independent, which in turn opens up previously unexplored analytical options. In line with [Stephenson's \(2007b\)](#) original proposal, we treat *find* as intensional for reasons of semantic composition (§3.2), though we are open to an extensional account provided it gets our data. At the same time, we will remain agnostic about the exact nature of propositional attitude it expresses. In many ways, our proposal treats *find* on a par with *like* and other psychological predicates, and as such is compatible with doxastic as well as non-doxastic views on *find*.

Finally, while we will not offer a semantics for *consider*, we will show that the linguistic profile of *find* cannot be explained if it is viewed merely as a more restricted/more subjective variant of *consider*. In §3.1, we show that many similarities between *find* and *consider* are due to the the previously unnoticed fact that, along with other predicates, they belong to the natural class we call ‘characterizing verbs’. And in §3.2–§3.5, we discuss a range of unique properties of *find* that set it apart from characterizing verbs, including *consider*, and that are not reducible to the strength of subjectivity. These data challenge the idea that *think*, *consider* and *find* form a scale and suggest a more interesting ontology.

We can now proceed to the empirical heart of the paper. The next section offers a guided tour of the linguistic behavior of *find* that firmly sets it apart from *think*, highlighting important parallels with other verb classes along the way.

with the same complements (*find delicious* vs. *consider delicious*) are due to a pragmatic competition driven by the Maximize Presupposition principle (first introduced by [Heim 1991](#); see overview in [Bade 2021](#)), according to which speakers opt for the presuppositionally strongest alternative. *Find* is taken to be a presuppositionally stronger alternative (it requires radical counterstance-contingency), so *consider* won't be chosen to express the same meaning as *find*, even though it is technically compatible with it. However, such pragmatic constraints can be overridden: *think*, often seen as a weaker alternative to *know*, can sometimes be used in contexts where *know* is appropriate. The fact that *consider* is consistently used in a way different from *find* suggests that (i) *consider* does not in fact express opinion, and that (ii) *consider* and *find* may not be scalemates (pace Kennedy and Willer).

3 Core data

While there is now substantial literature on *find*, this paper is the first attempt at a systematic investigation of its distinguished linguistic profile. This section examines the following properties, which were often not discussed at all or else received glancing attention. Importantly, our aspirations go beyond a mere description of what *find* is in contrast to *think* and other attitude predicates. We argue that its distinct properties, when taken together, point towards a complex semantic structure that has not been taken into account in current theories.

In §3.1, we extend Sæbø's (2009) puzzle on subject-predicate asymmetries under *find* to other predicates that take small clauses and argue that such small clauses are best understood as structured propositions.

In §3.2, we focus on the intensionality of *find*. Expanding on existing literature, we show that *find* does not allow referential opacity for noun phrases in its scope, which is unexpected if it is a doxastic predicate. At the same time, we provide arguments against purely extensional approaches.

In §3.3, we argue that *find* exhibits the signature behavior of emotive factives: it presupposes prejacent truth and the finder's belief (with respect to objective content) and asserts that the finder has an attitude about the prejacent (expressed by the subjective predicate in the embedded predicate position). This shows that *find* is a knowledge verb, not a belief verb.

In §3.4, we introduce a new moving part: aspectual class. There is increased awareness that aspectual properties of attitude predicates influence their overall semantics and can interact with the content of their complements. We demonstrate that *find* describes a complex change-of-state eventuality (from not having a judgment to having it), which in turn explains some of its differences with the stative *think*.

Finally, in §3.5 we turn to the acquaintance requirement of *find*, a property that is the subject of much ongoing debate. Our goal is to cement the idea that *find* semantically encodes experience with the small clause subject (but not, as in some previous accounts, with the proposition expressed by the small clause).

Many of our empirical observations are new and as such not discussed in previous work. To improve readability, we will provide an argumentative discussion of how different families of existing accounts handle our data based on a datapoint-by-datapoint rather than an account-by-account basis.

Given those preliminaries, we can now start our detailed exploration of what *find* does. We urge the reader to bear with us: we will emerge at the end with a clear understanding of the desiderata for a theory of *find*, which we will develop in §4.

3.1. Subject-predicate asymmetries and structured meaning

We start with an outstanding puzzle on subject-predicate asymmetries in *find*-complements. First pointed out by Sæbø (2009), it remains mostly unsolved in the existing literature. As the contrast in (10) illustrates, *find*-complements must contain a subjective expression in the main predicate position: (10a) with *delicious* is good, while (10b) with *Assamese* is infelicitous even though there is a subjective predicate elsewhere in the clause. *Think*, on the other hand, shows no such asymmetries (11). We will refer to such asymmetries as Sæbø's Puzzle throughout the paper.

(10) Sæbø's (2009)'s Puzzle

- a. ✓I find this Assamese tea delicious.
- b. #I find this delicious tea Assamese.

- (11) a. ✓I think that this Assamese tea is delicious.
 b. ✓I think that this delicious tea is Assamese.

Sæbø argues that subject-predicate asymmetries under *find* are due to a more general asymmetry in the denotations of predicative vs. attributive uses of subjective predicates (cf. (6)). While his specific implementation has been criticized in existing literature and we ourselves will argue that his account oversimplifies the many semantic contributions of *find*, we maintain the core intuition of his view. We will not make claims about matrix cases (*This tea is delicious*) and will focus our attention on the semantic composition of *find*-complements. Our central claim is that *find* belongs to a class verbs all of which give rise to the pattern in (10) and which can be identified by their syntax. The main take-home message is that syntax should be taken seriously as it can be instrumental in understanding meaning.

Much of the discussion of *find* has focused on the types of predicates it takes and how this allows us to differentiate *find* from *think* or *consider*. In this section, we want to shift attention to the nature of clauses embedded under *find*. While it is well-known that English subjective *find*, unlike its counterparts in, e.g., Norwegian (4), takes small clauses, this fact remained at the periphery of attention in semantics and philosophy of language. We aim to redress this gap. We will argue that *find* belongs to a hitherto undiscussed class of predicates which we call ‘characterizing verbs’. We show that syntactic data directly bear on the meaning of *find* and argue that small clauses under characterizing verbs are structured propositions that are semantically more complex than propositional complements of *think*: the subject and the predicate are composed differently so that the subject is singled out and the predicate describes some defining property of the subject. In adopting this treatment, we follow Bittner (1998a,b), who argues that complements of *consider* make both the subject and the predicate compositionally visible.⁹

In many ways, characterizing small clauses are about their subjects, which explains Sæbø’s (2009) Puzzle in a straightforward way: if *find* expresses an opinion about an entity in the embedded subject position of the complement, then the opinion must be contained within the predicate position. We conclude by hypothesizing about the information-structural properties of clauses embedded under *find* and other characterizing verbs and provide preliminary empirical support for the idea that all such clauses have a topic-comment structure with the verb upstairs imposing separate constraints on the embedded subject and the embedded predicate.

Small clauses Small clauses are non-finite tenseless complements whose predicate position cannot be filled by an inflected verb, but instead contains a non-inflected verb (e.g., so-called bare infinitives—verbs without *to*) or an adjectival, nominal or prepositional phrase (first introduced by Williams 1975; see overview in Cardinaletti & Guasti 1995). English has at least two types of small clauses, each with a distinct semantic and syntactic profile (Basilico 2003; Raposo & Uriagereka 1995).¹⁰

(12) PERCEPTION SMALL CLAUSES

Baby Toby heard/felt/noticed/observed/saw/watched [the Goblin king perform the magic dance].

9. Bittner is solving a different interpretational puzzle and only talks about *consider*, but we take her proposal as a confirmation of the idea that structured meaning has a place in understanding the denotations of objects attitude verbs take as arguments, and particularly for characterizing verbs as a class.

10. Following standard syntactic notation, square brackets in (12) and (13) demarcate the content of the complement clause.

(13) CHARACTERIZING SMALL CLAUSES

- a. Sisko considers [his position a challenge / an honor / a pleasure cruise / a treat].
- b. Nowadays many governments consider/deem/find/judge [universities dangerous and unprofitable].
- c. Formal semanticists often characterize/classify/define/describe/regard/see/treat/view [attitude predicates as a species of modals].

Complements in (12) describe the perception of an event and therefore exclude stative predicates, such as nouns and many adjectives (Carlson 1980; Higginbotham 1983; Jäger 2001). Complements in (13) describe the judgment about an entity; we will call predicates that take such complements ‘characterizing verbs’. Unlike the well-studied perception verbs featured in (12), these verbs have not been previously identified as a natural class (though see some initial remarks in Borkin 1973). While a detailed discussion of this new class is beyond the scope of this paper, we will delineate some of its semantic properties that are directly relevant in our quest to understand *find*.

Characterizing verbs Capitalizing on Basilico’s (2003) insights, we propose that small clauses under characterizing verbs have an asymmetric internal structure (there is ample evidence for syntactic differences between small clauses under perception vs. characterizing verbs). While Basilico focuses on *judge* and *consider* (and only mentions *find* once), our crucial innovations are to identify a whole class of verbs like this and to cement *find* as its member. These clauses are about their subjects, and their predicates express some defining quality of the subject (we stop at this intuitive description, and will not attempt to formally define relevant features that are responsible for non-trivial constraints imposed by each characterizing verb). *Find* requires a subjective property. *Consider*, as discussed in detail by Kennedy & Willer (2016, 2022); Muñoz (2019), requires a vague property that may be settled via conversational negotiation. *Deem*, *judge* or *regard as* seem similar, though more research is needed to finesse possible differences between those verbs. Many of characterizing verbs are happy with inherent properties that are often described in semantic literature as individual-level (cf. footnote 7), however, this doesn’t seem necessary. The important generalization is as follows: all characterizing verbs impose certain restrictions on embedded predicates (it should be a property suitable for judgment/definition) and their use is not licensed when an inappropriate predicate is inside their complement. Furthermore, they all exhibit the pattern discussed for *find*: the relevant property, whatever it is, must be contained in the predicate position of the embedded clause (Kennedy & Willer (2016) discuss this pattern for *consider*, but don’t generalize to other verbs). Thus, the sentences in (14a,b) are felicitous, while the sentences in (14c,d) are out even though they contain a suitable predicate elsewhere in the embedded clause.¹¹

- (14) a. ✓I consider/deem/judge this prime number fascinating.
b. ✓I regard/view this prime number as fascinating.
c. #I consider/deem/judge this fascinating number prime.
d. #I regard/view this fascinating number as prime.

We propose that characterizing small clauses in English are structured propositions: they have a bipartite structure with different parts made visible to semantic operators (see Krifka 1992, 2001; von Stechow 1991 for a general discussion of structured meaning in semantics).

11. In (14c,d) we focus on the issue of whether the number is fascinating; these sentences are felicitous if the issue in question is about prime-hood. See Kennedy & Willer (2022) for discussion of the context-dependence of these judgments with *consider*.

This move allows us to formulate the semantics of each verb in a way that it imposes constraints not just on the clause as a whole (as is often done for *find*), but separate restrictions on the embedded subject vs. predicate, which in turn resolves Sæbø's Puzzle in a principled way. This is a major distinction between the complements of characterizing verbs on the one hand vs. ordinary *that*-clauses on the other.¹² In §4.1, we discuss that such requirements fall under the general umbrella of semantic selection.

Information structure and the subjects of characterizing small clauses Our proposal for characterizing small clauses aspires to go beyond the technical solution to Sæbø's Puzzle. Instead, we view the puzzle as an instance of more general information-structural properties of such clauses. In the syntactic literature on small clauses, [Basilico \(2003\)](#) and [Raposo & Uriagereka \(1995\)](#) argue that characterizing small clauses in (12) differ from perception reports in (13) in that their subjects are topics that occupy a distinct left-peripheral position (higher than regular subjects). In semantic literature, various strands of research converge on the idea that structured meaning is the right analysis for a range constructions that single out the object of a predication and describe some property of that object, including topic-comment constructions and configurations expressing categorical judgment, with some authors explicitly arguing that the latter can be reduced to the former ([Jäger 2001](#); [Kuroda 1972, 1990](#); [Ladusaw 1994](#)). While a proper engagement with this diverse literature is beyond the scope of this paper, we hypothesize that characterizing small clauses do have a topic-comment structure and show that treating their subjects as topics may explain previously unnoticed constraints on their interpretation.¹³

Consider the contrasts below. Bare plurals in English are ambiguous between a generic and an existential reading ([Carlson 1980](#)), the former often called the strong reading (the strong-weak distinction is a descriptive term; see overview in [McNally 2020](#)). Characterizing verbs require the generic reading of bare plurals (17a) and block the existential one (17b), even though the predicate *tired* in and of itself is compatible with both readings in matrix (15) and embedded cases (16).¹⁴ Importantly, characterizing verbs allow overt existential quantifiers (18).

- (15) Students were tired.
- a. =Some students were tired. [✓∃]
 - b. =Students as a class were tired. [✓GEN]
- (16) The teacher thought that students were tired.
- a. = The teacher thought that some students were tired. [✓∃]
 - b. = The teacher thought that students in general were tired. [✓GEN]

12. It seems that *think* also used to have a life as a characterizing verb (such uses sound somewhat archaic right now): *Bismarck was different in all his creative capacities from his equals, so much so that they **thought him uncanny*** (*The New York Times* 1929, "Ludwig finds a real German statesman"). Furthermore, we suspect that *think of X as Y* (*I think of you as a friend*) can be a characterizing construction, which raises interesting new questions about the core lexical meaning of *think* when combined with complements of different types. However, we won't discuss those uses further and focus exclusively on *think* in combination with *that* clauses in this paper.

13. Topicality is a difficult notion that often resists precise formal characterization (see overview in [Tomioka 2021](#)). To this end, a proper discussion of the information-structural properties of characterizing clauses will take us too far afield and we will stick to a formal description of such clauses as structured propositions.

14. [Basilico \(2003:4\)](#) discusses data on bare plurals but incorrectly assumes that *consider* selects for predicates that block weak readings of their subjects. We make a weaker claim: *consider* and other characterizing verbs select for predicates that allow strong readings of their subjects; the requirement that the subject DP must have a strong reading comes from the embedded clause syntax (in line with Basilico's general claims).

- (17) The teacher considered/deemed/judged students tired.
- a. $\neq$ For some students, the teacher considered/deemed/judges them tired. [# $\exists$]
 - b. $=$ For students as a kind, the teacher considered/deemed/judged them tired. [✓GEN]
- (18) The teacher considered/deemed/judged some students tired.

Likewise, indefinite DPs with *a* or *one* are ambiguous between specific and non-specific readings (Fodor 1970a). Characterizing verbs require the specific reading and block the non-specific one (19a), even in those cases where the clause allows a non-specific reading otherwise (19b,c). Note that a non-specific reading of a different DP, *at least one*, is possible (19d).¹⁵

- (19) *Ormus Cama, a starting musician, is mixing a recording that involves many people singing in unison. He detects that someone is off-key, but with a sufficiently large number of voices and highly similar voices, it may be hard to know how many or to identify the person(s).*
- a. #Ormus considered/deemed/judged a/one singer off-key, but he doesn't know who it was.
 - b. ✓A singer/one singer was off-key, but Ormus doesn't know who it was.
 - c. ✓Ormus thought that a singer/one singer was off-key, but he doesn't know who it was.
 - d. ✓Ormus considered/deemed/judged at least one singer off-key, though he doesn't know who it was.

Without delving into details, we offer the following explanation to the pattern in (17) and (19a): subjects of characterizing clauses are topics of those clauses, and we are dealing with constraints on topicality. Existential readings of bare plurals and non-specific readings of indefinites are non-referential and cannot be topics in those constructions that require them (Cohen & Erteschik-Shir 2002, Erteschik-Shir forth., in line with Jäger 2001). If this explanation is on the right track, Sæbø's Puzzle follows from the general structure of characterizing small clauses: they form a topic-comment configuration, and the characterizing property must be a comment on the topic.¹⁶ Importantly, and this is the rationale behind this section, it is necessary to look at the whole class of characterizing verbs and not at *find* in isolation. While *find*-clauses exhibit the same pattern (only generic readings of bare plurals and specific readings of indefinites), it may be attributed to the properties of predicates that can occur under *find*. In English, subjective predicates that can occur under *find* are classified as individual-level (cf. footnote 7), and such predicates always require strong subjects even at the matrix level (Carlson 1980; Jäger 2001). The data on *find* can thus be explained without appealing to the small clause syntax. As we show above, other characterizing verbs can combine with predicates that do not have such restrictions (such as *tired* or *off-key*), which in turn demonstrates that interpretational constraints on noun phrases come from the presence of a characterizing verb and its syntactic requirements.

The key empirical takeaway is as follows: *find* differs from *think* in that only *find*, but not *think*,

15. Bouchard (2012) briefly notes that indefinite subjects under *find* must be specific, but does not connect that fact to the small clause syntax of characterizing predicates or other properties we discuss here.

16. In a somewhat related proposal, Kennedy & Willer (2016) attempt to derive Sæbø's Puzzle from a recursively-defined definition for issues raised by a proposition. They propose that quantified sentences raise issues about their nuclear scopes, hence the invisibility of subjective content in small clause subjects. However, they claim that quantified sentences raise issues only their nuclear scope, which would seem to predict that quantificational expressions could never be the focus of an answer to a question, contrary to fact.

gives rise to Sæbø’s Puzzle. We propose that this fact is directly linked to the small clause syntax of *find*-complements, a well-known property that has not been in any way exploited in semantic literature, and argue that *find*, along with *consider* and many other predicates, belongs to the class of characterizing verbs, which also give rise to Sæbø’s Puzzle. Drawing on several decades of research on English syntax and semantics, we propose that characterizing verbs take structured proposition arguments and speculate that small clause complements of characterizing verbs have a topic-comment structure, as evidenced by the interpretational constraints on the subjects of such clauses. Importantly, in this paper we are only concerned with English *find*. The surface syntax of its complements obviously differs from the syntax of *that*-clauses and we propose that this syntactic difference indicates a profound semantic difference as well. We do not know at present whether *find*-verbs across languages behave the same way or what other characterizing verbs there might be. However, given that subject-predicate asymmetries are present in Norwegian (Sæbø 2009), where *synes* only takes full clauses (4), we suspect that one can get substantial cross-linguistic mileage out of the structured meaning account we propose. We hope to see explorations of this idea in future research.

3.2. Referential transparency

Our next stop is the interpretation of referential noun phrases in the scope of *find*: we show that *find* does not allow true ‘de dicto’ readings for non-subjective content. While parts of the data have been brought up in previous literature, the overall pattern we discuss below has not been considered before in its entirety. We argue that this data is unexpected under several existing approaches, both intensional and extensional, and show that empirically *find* is most similar to emotive factives (§3.3). Expanding on Bouchard (2012) and Coppock (2018), in §4.4 we develop an account such that *find* is factive with respect to everything except subjective content in the main predicate position of the embedded clause, capitalizing on the asymmetry between the subjects and predicates of characterizing small clauses discussed in the previous section.

Approaches that treat *find* as a species of *think* predict that *find* will exhibit the pattern typical of intensional environments across the board, such as *think* or *say* (Farkas 1997; Musan 1997a; Percus 2000):

- expressions in the subject position of the embedded clause give rise to ‘de re’/‘de dicto’ ambiguities: they can be evaluated with respect to the local worlds or a higher world;¹⁷
- expressions in the main predicate position of the embedded clause must be read ‘de dicto’: they have to be evaluated with respect to the local worlds (though see Schwager 2009).

This prediction is not borne out: *find* does not allow true ‘de dicto’ readings for objective content and gives rise to the following asymmetric pattern (partially discussed by Bouchard 2012; Coppock 2018; Khoo 2026):

- both subjective and objective content in the subject position must be read ‘de re’ and are evaluated with respect to a higher world (‘de dicto’ is blocked);
- objective content in the predicate position must be evaluated with respect to both the local worlds and a higher world (‘de re’ + ‘de dicto’);¹⁸

17. Note that ‘de re’ is not an instance of the DP scoping out, as evidenced by the Double Vision cases discussed by Quine (1956) and much later work.

18. Note that this is not the same as the double-access reading (Abusch 1997), as temporality is not involved with *find*.

- subjective content in the predicate position must be read ‘de dicto’: it is evaluated with respect to the local worlds and reflects the finder’s perspective.

Subject position All content in the subject position of the embedded clause is interpreted ‘de re’ (20–22): subjective expressions reflect the speaker’s perspective, and objective expressions reflect facts at a higher world (=utterance world, w_0 , in the case of matrix uses of *find*). First of all, *find*-complements always talk about existing entities in the world of evaluation: unlike a corresponding sentence with *think* (20a), (20b) is felicitous only if the speaker endorses the existence of the under-the-bed monster. This is evidence that the embedded subject cannot be interpreted ‘de dicto’, relative only to the finder’s perspective.

- (20) a. My kids think that the monster under their bed is scary, ✓though of course it doesn’t exist.
b. My kids find the monster under their bed scary, #though of course it doesn’t exist.

The same kind of asymmetry holds when we consider cases where the speaker and the finder are acquainted with an existing object under different guises (Quine’s (1956) Double Vision cases), so the puzzle isn’t simply about existence, as in (20). (21) illustrates. The complement of *find* must contain the description true at a higher world (‘de re’) but not at the local worlds (‘de dicto’).

(21) The Scared Hiker Case

While hiking in the dark, your companion Kit became scared of what he thought was a mountain lion, not at all uncommon in the area. You saw, however, that it was a tree.

- a. ✓Kit thought that [a tree trunk]_{de re} / [a mountain lion]_{de dicto} was frightening.
b. ✓Kit found [a tree trunk]_{de re} frightening.
c. #Kit found [a mountain lion]_{de dicto} frightening.

Unlike *think* (21a), which allows both construals of the embedded subject, *find* only allows ‘de re’ (21b) and blocks ‘de dicto’ (21c). (22) illustrates the same pattern, but for subjective content. With *think* (21b), *frightening* can be about the attitude holder’s judgment if the host noun is interpreted ‘de dicto’ (we consider only autocentric, or attitude holder-oriented, readings of subjective predicates for now).¹⁹ However, a subjective expression inside the subject position of a *find*-complement must reflect the speaker’s perspective, which renders (22b) contradictory since it suggests that an entity is both frightening and harmless at the same time.

- (22) a. ✓Kit thought that [a frightening mountain lion]_{de dicto} was dangerous, even though it was just a harmless tree.
b. #Kit found [a frightening mountain lion]_{de re} dangerous, even though it was just a harmless tree.

Summing up, *find* blocks a ‘de dicto’ interpretation for embedded subjects: these DPs have to be evaluated with respect to a higher world.²⁰ This pattern presents a problem for several

19. Such data are discussed in detail by Anand & Korotkova (2022), who show that the interpretation of subjective expressions in intensional environments follow the same constraints as that of ordinary non-subjective predicates.

20. Interpretational constraints on small clause subjects can be explained syntactically if those noun phrases are part of the higher clause [Kit finds the tree trunk [frightening]] (e.g., base-generated there if we’re dealing with the so-called prolepsis (Lohninger et al. 2022; Yoon 2007) or moved if we’re dealing with subject-to-object raising, though Basilico (2003) explicitly argues against the latter option for *consider* and *judge*). In this case, ‘de dicto’ is not available since the noun phrase is not in the scope of an intensional operator introduced by *find*; with movement, one has to additionally assume that reconstructed nouns (those interpreted at their original, lower

intensional analyses of *find* (we discuss extensional approaches later). Thus, all accounts that treat *find* as a special version of *think* (Korotkova & Anand 2021; Stephenson 2007b; Bylinina 2017; Vardomskaya 2018) incorrectly predict that *find* would allow ‘de dicto’ readings such as (21c). Kennedy & Willer (2022) offer a more elaborate semantics, however, for our purposes their account treats *find* as *think* as far as referential opacity is concerned: since it doesn’t require the finder’s doxastic state with respect to factual issues to reflect the actual state of affairs, it incorrectly predicts that *find* would allow ‘de dicto’ readings for embedded subjects.

Predicate position *Find* patterns exactly like *think* in that subjective content is interpreted exclusively ‘de dicto’ in the main predicate position but not outside of it and that it must reflect the finder’s perspective. All other interpretations are blocked, as (23) illustrates (we assume that *depressing* and *uplifting* are antonyms that cannot hold of one object simultaneously).

(23) The Uplifting Movie Case 1

Pascal and you watched Moonage Daydream together; Brett Morgen’s documentary about David Bowie. You loved it, he didn’t.

- a. ✓ Pascal thought that the documentary was [depressing]_{de dicto}.
- b. ✓ Pascal thought that the [uplifting]_{de re} documentary was [depressing]_{de dicto}.
- c. #Pascal thought that the [depressing]_{de dicto} documentary was [uplifting]_{de re}.
- d. ✓ Pascal found the documentary [depressing]_{de dicto}.
- e. ✓ Pascal found the [uplifting]_{de re} documentary [depressing]_{de dicto}.
- f. #Pascal found the [depressing]_{de dicto} documentary [uplifting]_{de re}.

The pattern in (23) seems trivial and entirely expected if *find* is treated as a vanilla intensional operator. However, English subjective *find* can also take DP complements, provided that they contain as a subjective expression.²¹ In such cases, the host noun in the predicate position must be evaluated both with respect to a higher world and with respect to the local worlds (unlike nouns in the subject position that are only evaluated at a higher world). Consider (24), adapted from Anand & Korotkova (2022); we assume that *documentary* and *mockumentary* are antonyms.

(24) The Uplifting Movie Case 2

Pascal and you watched Moonage Daydream together; Brett Morgen’s documentary about David Bowie. You loved it, he didn’t. Additionally, you disagree about the nature of the movie—while you know it was a documentary, he mistakenly thinks that he watched a mockumentary.

- a. ✓ Pascal thought that [the uplifting documentary]_{de re} was [a depressing mockumentary]_{de dicto}.
- b. #Pascal found [the uplifting documentary]_{de re} [a depressing mockumentary]_{de dicto}.
- c. #Pascal found *Moonage Daydream* [a depressing mockumentary]_{de dicto}.
- d. #Pascal found *Moonage Daydream* a [depressing]_{de dicto} [documentary]_{de re}.

Think, a canonical intensional environment, requires the whole DP in the main predicate position of the embedded clause to be interpreted ‘de dicto’, i.e., with respect to the local worlds (24a). The most natural interpretation of subjective expressions inside this DP is an

position) cannot be read ‘de dicto’ (cf. Sharvit & Guerzoni 2003). We do not pursue this analysis here and simply note that for our purposes, it is also important to explain the interpretational pattern in (23) and (24), which is not amenable to the possible explanation sketched here.

21. This property is not universal, for example, German *finden* usually does not allow such configurations (Frühauf 2015; Reis 2013).

autocentric one, such that *depressing* reflects the attitude holder’s perspective. *Find* exhibits a different pattern. While subjective expressions inside the main predicate position must be read ‘de dicto’ (as discussed above for (23)), the host noun does not allow a pure ‘de dicto’ reading (24b,c): instead, it must be evaluated both with respect to the local worlds and the higher world.²² Finally, the host noun cannot be interpreted ‘de re’ either (24d). With respect to the content in predicate position of the embedded clause, the speaker and the finder must agree on facts. This part of the pattern is correctly predicted by Kennedy & Willer (2016), but not by Kennedy & Willer (2022) or several other intensional accounts that predict that *find* would pattern like *think* (Korotkova & Anand 2021; Stephenson 2007b; Bylinina 2017; Vardomsкая 2018).

In previous section, we argued that some properties of *find* are best understood in a broader context of characterizing verbs. However, referential transparency is not a general property of the class. As (25)–(26) show, several other characterizing verbs pattern like *think* and allow ‘de dicto’ in both subject and main predicate position of their complement (as Kennedy & Willer (2022) correctly predict for *consider*).

- (25) a. ✓Pascal considers/deems the uplifting documentary a depressing mockumentary.
b. ✓Pascal describes/treats *Moonage Dream* as a depressing mockumentary, even though it is a documentary.
- (26) a. ✓My kids consider/deem the monster under their bed scary, though of course it doesn’t exist.
b. ✓My kids describe/treat the monster under their bed as scary, even though of course it doesn’t exist.

Against extensional *find* We started with the idea that *find* is intensional and have shown that it does not pattern like other intensional operators, including *think*. Based in particular on data similar to (21), several authors treat the dissimilarities with *think* seriously and analyze *find* as an extensional operator (to the best of our knowledge, the first approach of this sort is due to Sæbø 2009; it was later defended by Franzén 2020, 2021 and Khoo 2026). Prima facie, it solves the problems discussed so far: if *find* does not introduce intensional quantification, noun phrases in its scope will be always interpreted relative to the world of evaluation. However, extensional views are not unproblematic either. Consider (27) and (28).

- (27) #The uplifting documentary is depressing.
(adapted from Anand & Korotkova 2022:623)
- (28) a. ✓Pascal thought that [the uplifting documentary]_{de re} was [depressing]_{de dicto}. [=23b]
b. ✓Pascal found [the uplifting documentary]_{de re} [depressing]_{de dicto}. [=23e]

(27) contains two opposing predicates and is a contradiction (unless used with special intonation or intensional markers like *actually*, see Anand & Korotkova 2022:623, fn.4). On the other hand, (28a) is felicitous. Anand and Korotkova argue that subjective predicates do not differ from non-subjective predicates in terms of their interaction with intensional operators and that (28a) is felicitous for the same reason as (29b), due to ‘de re’/‘de dicto’ ambiguities that ameliorate contradictions and are only possible in intensional environments, such as

22. Note that the infelicity of (24b) and (24c) is semantic, not syntactic, as *find* in principle allows two DPs inside its complement: *Sisko found the young cadet an exciting addition to his team*.

complements of *think* (29b) but not matrix clauses (29a).²³

- (29) a. #The deciduous tree is evergreen.
 b. ✓ Mordecai believes that [the deciduous tree]_{de re} is [evergreen]_{de dicto}.
 (Anand & Korotkova 2022:625)

If *find* is intensional, the felicity of (28b) is amenable to the same explanation as (28a) and (29b), without overgenerating for (27). However, the proponents of extensional *find* cannot appeal to this logic. If *find* does not shift worlds, the felicity of (28b) must be due to a mechanism separate from intensional quantification that allows two subjective predicates to be evaluated relative to different perspectives under *find*. But if this mechanism is not tied to the evaluation at separate worlds, it is not clear why (27) is infelicitous. While we are in principle open to the idea that *find* can be analyzed as an extensional operator, in this paper we will stick to our intensional guns for reasons of semantic composition and treat *find* as a quantifier over possible worlds, in line with the original proposal in (Stephenson 2007a) and much further research. See also Crespo & Veltman (2019) for additional arguments in favor of intensional *find*.²⁴ This is not to say that an extensional semantics for *find* is unviable altogether. However, if one adopts such a view, it is important to make sure that it is coupled with a theory of subjective expressions that correctly predicts the infelicity of (27). In this paper, we do not develop such a theory and therefore use an intensional analysis.

To recapitulate, *find* exhibits a peculiar pattern with respect to the interpretation of referential noun phrases in its scope, exhibiting sensitivity not only to the subjective/objective content (noted in existing literature) but also to the split between subject/main predicate position. All content in the subject position is evaluated with respect to a higher world ('de re') and does not have to reflect the finder's information state. Objective content in the predicate position is evaluated with respect to a higher world and the local worlds ('de re' + 'de dicto'), reflecting both the speaker's and the finder's information state. We will derive the asymmetry between subject vs. predicate position from small clause syntax.

3.3. Factivity

The previous section has demonstrated that *find* does not behave like a typical intensional environment (as would be expected if it were a specialized version of *think*), nor that it is easily amenable to a fully extensional account. This section shows that *find* is a knowledge verb, not a belief verb, and that it is most similar to emotive factives.

Bouchard (2012) and Coppock (2018) assume a type distinction between subjective vs. objective content that is visible to semantic operators and argue that all non-subjective content must project and thus escape the scope of the intensional operator introduced by *find*. Though neither Bouchard nor Coppock use this term, their accounts treat *find* as a factive predicate with respect to objective content. We sharpen this essentially correct intuition through

23. Deciduous trees shed foliage in the fall, while evergreen ones don't, so those properties are complementary antonyms.

24. Another argument for intensional *find* comes from sequence-of-tense phenomena. In those languages that have relative tenses (those that are no longer anchored to the speech time under attitudinal operators; Ogihara & Sharvit 2012) and where *find* takes full clauses, *find*-verbs can trigger sequence-of-tense (cf. Sæbø 2009 on Norwegian). Tenses are standardly bundled together with worlds in indices of evaluation (Musan 1997a), so any extensional account of *find* has challenges explaining sequence-of-tense under *find*, as Sæbø himself notes; such data are entirely expected under most intensional accounts, on the other hand. While we believe that this is a promising line of defense for intensional *find* across the board, our concern is English. And since small clauses have no tense, we cannot check the relevant data.

an explicit comparison between *find* and canonical factive predicates, focusing especially on emotive factives. The very distinction between cognitive factives (*discover*, *find out*, *know*, *realize*) and emotive factives (*be amused*, *glad*, *sad*, *surprised*, *mad*) has a long history; see detailed overviews in (Anand & Hacquard 2014; Djärv 2019). Traditionally these verbs have been treated as lexically presupposing the truth of their complements (Karttunen 1971, 1973; Kiparsky & Kiparsky 1970), though recent research advocates a more nuanced view on presupposition triggering (Scontras & Tonhauser 2026). Our main focus is the fact that *find* gives rise to certain entailments that put in squarely in the knowledge group, an observation that has not been made in the previous literature. The exact mechanism by which those entailments are backgrounded is less important, so for now we will treat them as lexical presuppositions (we come back to it in §5.1).

First of all, *find* patterns exactly like factive predicates with respect to the interpretation of embedded subjects. As (30) illustrates, both emotive factives like *regret* and cognitive factives like *know* block a ‘de dicto’ reading of embedded subjects (though this isn’t explicitly discussed in the literature, it follows from most accounts of factivity). All content, subjective and objective, is read ‘de re’.

- (30) a. Pascal discovered/found out/is surprised/learned/knows/regrets that [the uplifting documentary about Bowie]_{de re} won a Grammy award, ✓ even though he mistakenly thinks that it is a depressing mockumentary.
 b. Pascal discovered/found out/is surprised/learned/knows/regrets that [the depressing mockumentary about Bowie]_{de dicto} won a Grammy award, # even though it actually is an uplifting documentary.

Importantly, in many other respects *find* is close to emotive factives (*be sad*, *be surprised*, *regret*) which presuppose truth at a higher world ($\approx$ the speaker’s belief) and truth at the local worlds ($\approx$ the attitude subject’s belief), and assert the subject’s psychological state about the complement proposition. (31) and (32) illustrate these basic properties for *surprise* using the standard contradiction test to diagnose entailment and projection under negation as a test for presupposition (where w_0 is the evaluation world).

- (31) Pascal is surprised that *Moonage Daydream* is a documentary.
 a. ...#even though it is a mockumentary.
 → *Moonage Daydream* is a documentary in w_0 .
 b. ...#even though he thinks it is a mockumentary.
 → Pascal thinks that *Moonage Daydream* is a documentary.
 c. ...#even though he expected this.
 → Pascal didn’t know this before.
- (32) Pascal isn’t surprised that *Moonage Daydream* is a documentary.
 a. ...#even though it is a mockumentary.
 → *Moonage Daydream* is a documentary in w_0 .
 b. ...#even though he thinks it is a mockumentary.
 → Pascal thinks that *Moonage Daydream* is a documentary.
 c. ...✓he actually expected this.
 ↯ Pascal didn’t know this before.

We argue that *find* exhibits a very similar pattern: it presupposes that the main predicate description is true at a higher world as well as the local worlds and asserts that the finder has a propositional attitude about the entity (expressed by the subjective predicate). As discussed

in detail in §3.2, the pattern with embedded subjects is different since *find* takes structured propositions.

- (33) Pascal found *Moonage Daydream* an uplifting documentary.
- a. ...#even though it is a mockumentary.
→ *Moonage Daydream* is a documentary in w_0 .
 - b. ...#even though he thinks it is a mockumentary.
→ Pascal thinks that *Moonage Daydream* is a documentary.
 - c. ...# even though he also found it depressing.
→ Pascal judged the movie uplifting.
- (34) Pascal didn't find *Moonage Daydream* an uplifting documentary.
- a. ...#even though it is a mockumentary.
→ *Moonage Daydream* is a documentary in w_0 .
 - b. ...#even though he thinks it is a mockumentary.
→ Pascal thinks that *Moonage Daydream* is a documentary.
 - c. ...✓ actually he found it depressing.
↯ Pascal judged the movie uplifting.

As for emotive factives (31b,32b), the inference about Pascal's belief that *Moonage Daydream* is a documentary is presupposed (33b,34b). These data are problematic for the proposals of Bouchard (2012) and Coppock (2018), which treat *find* as a type of cognitive factive (ignoring the subjective content, which is asserted). However, since they mainly look at *find*-complements that don't contain DPs in the predicate position, they don't predict that the subject must be aware of the entity's identity (they predict that *find* would allow 'de dicto' readings in the main predicate position). We discuss such data in §4.4. The main idea is that the noun, when present, must be part of the finder's mental representation of the entity in question in addition to being an accurate description of that entity at a higher world.²⁵

The key difference between *find* and canonical factives is the interpretation of subjective content in main predicate position. With factives, subjective content may be non-autocentric and must be presupposed. With *find*, it must be autocentric and there is no entailment or presupposition that this content holds at w_0 . We postpone the discussion of obligatory autocentricity of subjective content under *find* until §5.1 and focus on the entailments about truth for now. Compare (35) and (36).

- (35) a. Pascal is surprised that *Moonage Daydream* is uplifting # even though it is depressing.
→ *Moonage Daydream* is uplifting at w_0 .
- b. Pascal isn't surprised that *Moonage Daydream* is uplifting # even though/because it is depressing.
→ *Moonage Daydream* is uplifting at w_0 .

25. Bouchard (2012) explicitly argues for the claim that *find* does not require the finder to commit to the non-subjective content in *find*-complements. His argument is based on the following example (minimally modified from his *John finds that you bought a nice car* so that it involves a small clause complement):

(i) John finds your car nice.

Bouchard observes that the finder may not be aware of who owns the car, and uses that to argue that the finder need not commit to non-subjective content in the embedded clause. However, given our discussion in §3.2, we can see this as an instance of 'de re' interpretation of non-main predicate content.

- (36) a. Pascal found *Moonage Daydream* uplifting ✓ even though it is depressing.
 ↗ *Moonage Daydream* is uplifting at w_0 .
 b. Pascal didn't find *Moonage Daydream* uplifting ✓ because it is depressing.
 ↗ *Moonage Daydream* is uplifting at w_0 .

One could still argue that *find* is a kind of emotive factive such that its affective content is about the subjective predicate (in line with, e.g., Franzén 2020, 2021): just like the speaker needn't share Pascal's emotions about the complement proposition with *be surprised* (it is always anchored to the attitude holder), the speaker needn't share Pascal's subjective evaluation with *find* (it is always anchored to the finder). Developing such a theory would require a proper understanding of subjective predicates in the scope of factive predicates, which we won't do in this paper. Our own proposal remains agnostic about the nature of the attitude asserted by *find* and is compatible both with doxastic and non-doxastic approaches. Our core observation is that in addition to asserting a judgment, *find* also presupposes that the finder is aware of the objective content inside the main predicate position. This fact has remained unnoticed in previous work and as such is not always predicted by existing approaches.

3.4. Aspectual class

Aspectual class (or Aktionsart, the type of eventuality denoted by the verb/verb phrase) is a classic way to group verbs into linguistically useful categories (the first classification was introduced by Vendler 1957; see Filip 2020 for an overview). With few exceptions, there is little work on the aspectual properties of attitude predicates (Anand & Korotkova 2024; Bondarenko et al. 2024; Korotkova & Anand 2024; Özyıldız 2021; Rawlins & White 2019; Zuchewicz 2020). Existing research shows that lexical aspect interacts with other facets of attitudinal meaning and is therefore important in understanding the semantics of the base predicate. In this section, we bring to the table a previously unnoticed fact: unlike *think*, *find* is a classic achievement predicate that denotes an instantaneous change-of-state eventuality and patterns like *strike* or *occur*. Unlike simple states that merely hold at an interval, achievements denote the point of transition from not being in the state expressed by the prejacent to being in that state and often linguistically encode additional parts of this complex underlying structure (Heyde-Zybatov 2008; Kearns 2003; Gyarmathy 2015; Piñón 1997; Piñón 2008). In §4 we exploit the ontology associated with achievement predicates to derive in a principled way the presuppositional profile of *find* discussed in the previous section and the acquaintance requirement we introduce in §3.5.

Find is not a state We show that *find* fails the following classic diagnostics for states (Dowty 1979; Rothstein 2004): compatibility with non-habitual simple present (37) and with *for*-adverbials (38). We compare it to *think*, a classic state (when combined with declarative complements; Özyıldız 2021). We further show that unlike achievements with a more involved semantics, *find* makes no obligatory reference to a preparatory process, as evidenced by its incompatibility with temporal frame adverbials (39), with the progressive (40) or with the scalar reading of *almost* (41). Note that we deliberately limit our attention to cases where the embedded clause subject is a singular definite individual with a very limited lifespan (a shooting star). The goal of this set-up is to ensure that any potential 'protracted event' effects (e.g., habitual or iterative readings) come from the base verb and not from the plurality introduced by event participants.

The Fleeting Beauty Case:

We're watching an unusually bright meteor shower, with several fireballs and even one purple star.

- (37) Ongoing, non-habitual simple present: ✓*think*, #*find*
- a. ✓Right now, I think that the the purple shooting star is dazzling/exquisite/glorious.
 - b. #Right now, I find the purple shooting star dazzling/exquisite/glorious.
- (38) *For*-adverbials: ✓*think*, #*find*
- a. For a minute/moment/split second, I thought that the purple shooting star was disappointing/distressing/ominous, but then came to adore it.
 - b. #For a minute/moment/split second, I found the purple shooting star disappointing/distressing/ominous, but then came to adore it.

The ongoing, happening-right-now reading of the simple present diagnoses states: all other eventualities get a habitual reading (*I ride the bike to work* can only mean that this is what I do in general, but not at the moment of speaking). If *find* were a state, we would expect it to pattern like *think* (37a) and many other doxastic predicates (*believe*, *be aware*, *understand*). However, even though *find* is in principle compatible with the simple present, it cannot be modified by *right now* to describe the situation unfolding at the moment of utterance (37b). *For*-adverbials diagnose eventualities that have duration, including ordinary states like *be tired* and special types of achievements that make their derived state accessible for semantic operators (e.g. *she went away for a day* in which the being away lasted a day, not the going away; such eventualities are sometimes called ‘left-boundary’ achievements, Heyde-Zybatov 2008). *Think* is felicitous with *for*-adverbials (38a), but *find* is more tricky. As we discuss in §4.3, it can be acceptable when certain conditions are met, but the single event of evaluating a short-lived entity is bad even with *for*-adverbials denoting infinitesimal intervals (38b), which demonstrates that finding lacks duration and is therefore not a state. Based on the data above, we conclude that *find* is an achievement (using the standard feature designation, we could describe it as +telic, +dynamic, -durative).

***Find* is a simple achievement** Recent literature establishes that achievements come in several varieties, including those that have a preparatory process and thus resemble accomplishments in terms of their event structure (a classic case will be *Lucy Walker reached the summit of the Matterhorn*; see detailed discussion and references in Gyarmathy 2015). Below we establish that subjective *find* is a truly instantaneous achievement. Thus, it is incompatible with *in*-adverbials even on the prospective reading synonymous with *after*-adverbials (Piñón 1997:278): such adverbials signal the presence of a linguistically accessible pre-state that ended prior to the culmination, and their infelicity with *find* (39) shows that *find* does not encode such a state. *Find* is generally bad with the ongoing process interpretation of the progressive, which shows that finding lacks stages (40), unlike, for example, the event of writing a paper (a classic accomplishment).²⁶ Finally, to the extent that *find* is compatible with *almost* (41), it only gives rise to the counterfactual reading (something could have happened, but didn’t) and not to the scalar reading (a process was initiated but not completed). The latter is possible only with those

26. A cursory corpus search returns very few examples of subjective *find* with the progressive: *I was finding the test so irritating and boring* (<https://statmodeling.stat.columbia.edu/2010/10/18/face-blind/>), so we will not consider them here. Additionally, (40) can be felicitous in the iterative reading of the progressive when the finder re-experiences the shooting star; see related discussion in §4.3.

eventualities that have a process with an endpoint (see references and discussion in Grob 2022), so the lack of this reading with *find* shows that there is no such process.

- (39) *In*-adverbials: #*find*
 #In a minute/fraction of a second, I found the purple shooting star dazzling/glorious/ominous.
- (40) The progressive: #*find*
 #I was finding the purple shooting star dazzling/glorious/ominous.
 #The more I looked at it, the more I was finding the purple shooting star dazzling/glorious/ominous.
- (41) *Almost*: ✓counterfactual, #scalar
 I never hated him, though. I almost found him very charming and handsome.²⁷
 = I could have found him charming and handsome, but didn't. [counterfactual]
 ≠ I started finding him charming and handsome, but didn't finish. [scalar]

Taken together, the data in (37)–(41) show that the basic semantics of subjective *find* refers to the transition from not having an attitude towards having it.²⁸ Note, however, that this is not a general property of characterizing verbs, many of which are states (42).²⁹

- (42) a. Right now, I consider the purple shooting star ominous.
 b. Right now, I regard you as a friend.

To recapitulate, *find*, but not *think*, is an achievement predicate. This is not a mere comment on the linguistic profile of *find* that adds to a collection of properties differentiating it from *think*. Instead, we argue that the aspectual data serve as a window on the complex conceptual structure of the eventuality of finding. *Think* expresses being in a state that is decided on the propositional content of its complement. *Find* expresses a change of state that results in the finder's having an attitude expressed by the complement. Many telic predicates have an underlying causal structure that includes the trigger for the change of state in the lexical semantics of the verb (Nadathur 2023; Nadathur & Siegal 2025). We postulate the same general structure for *find* in §4. We start by arguing in §3.5 that *find* lexically encodes the presupposed event of experience that in turn leads to the judgment asserted by the predicate.

To the best of our knowledge, the only discussion of the aspect of subjective *find* is due to Vardomskaya (2018:126-127), who incorrectly classifies it as a state. We believe that this observation, mistaken though it may be, underpins much research on *find* in comparison to *think* and is ultimately rooted in the conspiracy of factors that allow what looks like a stative *find* with simple present (1b, repeated below as (43)):

- (43) Q finds humans entertaining. [=1b]

Find requires experience (§3.5), and simple present (which can only combine with non-states with the help of a separate operator) is felicitous with *find* when the experience is repeatable, often facilitated by generic or plural embedded subjects. In this, but only in this reading, *find* is compatible with *for*-adverbials. We discuss this interpretation in detail in §4.3.

27. <https://www.wattpad.com/387399438-kenickie%27s-girl-under-heavy-editing-and>

28. Other uses of *find* are also achievements, though not necessarily of the same type; for example, *find* + DP (*find the treasure*) combines with *in*-adverbials, may have a preparatory process (Heyde-Zybatov 2008; Piñon 2008), admits the progressive, and allows a scalar interpretation of *almost*.

29. *Consider* is an activity when combined with interrogative complements, similarly to how *think* behaves in English (Özyıldız 2021) or Georgian (Bondarenko et al. 2024).

3.5. Acquaintance requirement

In this section, we defend against existing contenders the view that *find* semantically encodes an Acquaintance Inference (the AI; term due to [Ninan 2014](#)), the requirement that the finder has some experience with the small clause subject. Unlike other properties discussed in this section, the acquaintance requirement has received substantial attention in the literature. The idea that *find* encodes a kind of sensory experience first appears in [Borkin \(1973\)](#) and later in [Stephenson \(2007b\)](#).³⁰ The debate is ongoing: some strands of subsequent research challenge experiential *find* ([Bylinina 2017](#); [Kennedy & Willer 2016, 2022](#); [Muñoz 2019](#)), some endorse it wholeheartedly ([Anand & Korotkova 2018](#); [Korotkova & Anand 2021](#); [Vardomsкая 2018](#)), while some others remain non-committal ([Bouchard 2012](#); [Coppock 2018](#); [Sæbø 2009](#)). Drawing on previous literature as well as novel empirical contrasts, in §4.2 we couch the AI of *find* as a presupposed event of experience with the small clause subject that causes the resulting judgment asserted by *find* (we touch upon those issues also in §5.1). Along the way we demonstrate why this account is superior to some other analytical options present on the market.³¹

Find and AI-triggering predicates in its scope [Borkin \(1973\)](#) first proposed that *find* encodes sensory experience. [Stephenson \(2007b\)](#) argued that this requirement is part of the lexical semantics of *find*, which is, in turn, another property that distinguishes it from *think*:

- (44) $\llbracket \text{find} \rrbracket^w = \lambda x \lambda p. \forall w' \in \text{DOX}(x, w)[p(w') \text{ and this is caused by } x\text{'s having sensory experience of } p \text{ in } w]$, defined iff p is subjective.

(adapted from [Stephenson 2007b:61](#))

A number of authors argue that *find* merely selects for predicates that have their own acquaintance inference and therefore does not in and of itself have experiential semantics ([Bylinina 2017](#); [Muñoz 2019](#)). (45) illustrates this basic pattern: *find* requires experience in (45a) in line with unmodified uses of *delicious* (45b).³² Note that in all cases below, we carefully construct our examples to create the referential reading of the definite DP and avoid the kind reading.

- (45) a. I found the puerh you just brewed delicious, #though I haven't tried it.
 → Speaker has experience with the tea.
 b. The puerh you just brewed is delicious, #though I haven't tried it.
 → Speaker has experience with the tea.

The AI of subjective expressions such as *delicious* gives rise to a perplexing projection pattern ([Anand & Korotkova 2018](#); [Ninan 2014, 2020](#)). Like classic presuppositions, it projects under negation, but unlike classic presuppositions, it is blocked by epistemic modals such as *must*. Importantly, it may be blocked by attitude predicates such as *think* (though it does not have to be).³³ The accounts in [Bylinina \(2017\)](#) and [Muñoz \(2019\)](#) treat *find* as a species of

30. [Ducrot \(1975\)](#) discusses similar data for French *trouver*.

31. [van Wijnbergen-Huitink \(2016\)](#) argues that *find* cannot encode experience since it happily allows abstract properties, such as moral predicates, in its complements: *I found his actions immoral*. This is unproblematic for the view we advocate in §4.2 since we require that the finder experienced the embedded clause subject without specifying that the manner of experience involves a sensory encounter.

32. As discussed by [Anand & Korotkova \(2018\)](#), *delicious* only signals that someone, but not necessarily the speaker, has relevant experience. *Find* is more stringent: it requires that the finder is the experiencer, a fact we come back to in §5.1.

33. Some authors incorrectly argue that *believe* ([Stephenson 2007b:63](#)) and even *think* encode 'indirectness': instead of simply allowing the local accommodation of the AI of *delicious* in its scope (in line with the behavior

think, which predicts that *find* should behave like *think* and thus be able to prevent the AI of *delicious* from projecting to the matrix level (46a). This prediction is incorrect: *find* gives rise to an experience inference both in the positive form and under negation (46b).

- (46) a. She $\left\{ \begin{array}{c} \text{thought} \\ \text{didn't think} \end{array} \right\}$ that the puerh you just brewed was delicious, ✓even though/because she hasn't tried it.
 b. She $\left\{ \begin{array}{c} \text{found} \\ \text{didn't find} \end{array} \right\}$ the puerh you just brewed delicious, #even though/because she hasn't tried it.

One might think that the factivity of *find* can explain the pattern in (46b). But factivity will simply require that the puerh (at the world of evaluation) has the property of being delicious according to the finder; it will not require experience by the finder (any more than *know* does). Even if *find* selects for predicates that are experiential, as Bylinina (2017) and Muñoz (2019) suggest, it will not be enough, since selection in and of itself doesn't require that the experience is actually instantiated at the world of evaluation. The data in (46b) are left unexplained.

Experiential *find* As discussed already by Solt (2018) and especially by Korotkova & Anand (2021), *find* gives rise to an acquaintance inference even with those predicates that have no such requirements on their own, such as *bumpy* or *authentic*. In line with those authors, we believe that experience effects in those cases are due to *find*. Consider the following contrasts with *challenging* (47), based on Star Trek Universe.

Jake Sisko at the end of *Deep Space 9* has amassed more knowledge as to what a career in Starfleet is like compared to B'Elanna Torres at the beginning of *Voyager*, and yet only the latter can appropriately make a *find*-claim, since she actually made an attempt at having a Starfleet career.

(47) The Challenging Career case

Context 1 (firsthand experience): *B'Elanna Torres is a character in Star Trek Voyager. At the beginning of the series, we see her as a Starfleet Academy dropout who preferred joining the renegade Marquis over continuing a career in Starfleet.*

Context 2 (observation, lack of firsthand experience): *Jake Sisko is a character in Star Trek Deep Space 9. He is surrounded by Starfleet officers whose life has been changed in many ways through their career choices, including death (his mother), injury (his best friend Nog) and tough decisions during the Dominion wars (his high-ranking officer father). He never served himself.*

of intensional operators), these verbs are argued to require that the attitude subject lacks sensory evidence for the complement proposition. This is a misconception. *Think* often creates the effect of uncertainty due to its pragmatic competition with *know* (often explained along the Maximize Presupposition lines such that *know* is a presuppositionally stronger alternative to *think*), however, *think* does not encode uncertainty or indirectness, and is in fact compatible with sensory experience.

- (i) My mother knitted me a green sweater and I put it on. I thought it was pretty.

(Tove Ditlevsen, *Childhood, Youth, Dependency*)

The relevant contrast here is with doxastic predicates that encode inference (*assume*, *infer*, *presume*): they are semantically incompatible with sensory experience for the prejacent and therefore, unlike *think*, always block the AI of *delicious* in their scope.

- (ii) I've just tried the tea you brewed. I ✓think/#assume that it is delicious.

- a. A career in Starfleet is challenging.
Context 1:✓, Context 2:✓
- b. I find a career in Starfleet is challenging.
Context 1:✓, Context 2:#
- c. I consider a career in Starfleet is challenging.
Context 1:✓, Context 2:✓
- d. ✓B'Elanna Torres finds/found a career in Starfleet too challenging.
- e. #Jake Sisko finds/found a career in Starfleet too challenging.

Challenging, unlike *delicious*, does not trigger an AI of its own (47a), but *find*-complements containing it do (47b,d,e).³⁴ The proposals in Bylinina (2017) and Muñoz (2019) incorrectly predict that *find* will not combine with *challenging* to begin with. Below we show that such data are problematic for other approaches that explicitly argue against an experiential *find*: the accounts in Kennedy & Willer (2022) and Khoo (2026).

Kennedy & Willer (2022) argue that what we call the acquaintance inference arises through a combination of the following properties. First, both *find* and *consider* give rise to familiarity inferences such that the attitude holder must be in possession of enough facts to make the judgment. Additionally, some predicates that occur under *find* and *consider* require *affective grounding*, which often amounts to sensory experience and is meant to explain the AI in cases like (45a). Finally, recall from §2.3 that both *find* and *consider* are argued to presuppose the counterstance-contingency of their complements. More specifically, *find*-complements are argued to be radically counterstance-contingent ($\approx$ genuinely subjective matters that can't be settled in conversation). We argue that this sophisticated account is still insufficient to explain the restrictions *find* imposes. Consider the following contrast:

(48) The Stereotypical Reasoning case

There is a new academic boycott. You don't believe in such measures, as they only breed hatred and polarization and ultimately don't target the real perpetrators.

- a. ✓This new academic boycott is meaningless and reprehensible. I've no idea what exactly it amounts to, but I don't need to.
- b. ✓I consider this new academic boycott meaningless and reprehensible. I've no idea what exactly it amounts to, but I don't need to.
- c. #I find this new academic boycott meaningless and reprehensible. I've no idea what exactly it amounts to, but I don't need to.
- d. #I abhor/detest/dislike/hate this new academic boycott. I've no idea what exactly it amounts to, but I don't need to.

34. One may wonder if *challenging* is an AI-triggering predicate that, unlike *delicious*, readily admits non-autocentric uses when unembedded but must be autocentric under *find*, which then creates an impression that the AI in (47b) is due to the presence of *find*. However, unlike *delicious*, which always triggers an AI when unembedded, both in its autocentric and non-autocentric uses alike (Anand & Korotkova 2018), *challenging* has purely dispositional uses such that the predicate holds of an entity even if the property has never been actualized (a hallmark of dispositional predicates across the board; Choi & Fara 2021).

(i) a. Obligatory AI with *delicious*:

This new cat food is delicious/tasty, #but no cat has ever tried it yet.

(based on Anand & Korotkova 2018)

b. Lack of AI with *challenging*:

A new position of the ambassador to the Dominion has opened up. It is very challenging, ✓even though nobody has held it yet.

(48) is a case of stereotypical reasoning: the speaker makes a moral judgment based on their general attitude towards boycotting. This type of reasoning is complementary with reasoning based on experience, and there are various linguistic expressions, including *bona fide* evidentials, that are sensitive to this distinction and require reasoning based on experience (Krawczyk 2012; Korotkova forth.). Neither *meaningless* nor *reprehensible* have an acquaintance inference, so the matrix case is felicitous (48a). Embedding under *consider* is equally felicitous (48b). However, *find* is not licensed in inferential cases, including stereotypical reasoning. In this respect, *find* patterns like canonical psychological predicates, which require acquaintance of the experiencer subject with the object (Cohen & Erteschik-Shir 2002) and ban stereotypical reasoning (48d), a resonance we will return to throughout the paper. Stojanovic & McNally (2023) discuss other cases where moral predicates (which do not require experience on their own, unlike taste or aesthetic predicates; Soria-Ruiz 2024) become experiential under *find*, but not *consider*. We treat such data as further support for experiential *find*.

We argue that any non-experiential account of *find* has difficulties in explaining the infelicity of (48c). For example, Kennedy & Willer (2022) predict (48c) to be felicitous. Thus, having a general opinion about boycotts is compatible with radical counterstance-contingency, as this is not something that may be settled via conversational negotiation (even if we agree on what counts as *reprehensible* and *meaningless* in general, we can still disagree about the extensions of those predicates). Furthermore, the infelicity of (48c) cannot be due to the affective grounding of the embedded predicate, given the felicity of (48a). Finally, if (48c) with *find* is ruled out due to the insufficient familiarity with this specific instance of boycotting, then (48b) with *consider* is also predicted to be infelicitous as *find* and *consider* are argued to have the same familiarity requirements. To conclude, the combination of familiarity inferences, radical counterstance-contingency and affective grounding does not explain why (48c), but not (48b), is infelicitous. For our present purposes, the account in Khoo (2026) works similarly. It treats *find* as encoding a particular emotional or psychological response in the finder, which differentiates it from *think* which encodes a belief. However, a strong reactive response needn't be grounded in experience (as Khoo explicitly argues), which in turn incorrectly predicts (48c) to be felicitous. We conclude that non-experiential accounts of *find* cannot capture its infelicity in stereotypical reasoning scenarios.

Summing up, we have presented a range of arguments in favor of experiential *find* over various other approaches and in the rest of the paper, we will assume that experience effects discussed thus far are due to the conventional meaning of *find*. We provide an additional argument for this semantic view in §4.3.

Status of the AI with *find* We hope to have convinced the reader by now that *find* is best understood as having experiential semantics. Note that even though *find* does not behave like classic attitude predicates, it patterns exactly like canonical psychological predicates such as *hate*, *like* or *love*, which also encode opinion based on experience (a fact reflected in their argument structure). Importantly, in each case the experience is conversationally backgrounded: it projects under entailment-cancelling operators and is never the main point of an utterance.

First, as discussed by Anand & Korotkova (2018) for *find* and Anand & Korotkova (2022) for *like*, the AI in each case behaves like a classic presupposition in terms of projection: it is entailed by unmodified affirmative sentences (49a,50a), with negation (49b,50b) and with epistemic modals (49c,50c).

- (49) a. Henk found the mapo tofu you made last night authentic, # though he hasn't tried it.
→ Henk has experience with the tofu.
- b. Henk didn't find the mapo tofu you made last night authentic, # though he hasn't tried it.
→ Henk has experience with the tofu.
- c. Henk must have found/probably found the mapo tofu you made last night authentic, # though he hasn't tried it.
→ Henk has experience with the tofu.
- (50) a. Henk liked the mapo tofu you made last night, # though he hasn't tried it.
→ Henk has experience with the tofu.
- b. Henk didn't like the mapo tofu you made last night, # though he hasn't tried it.
→ Henk has experience with the tofu.
- c. Henk must have liked/probably liked the mapo tofu you made last night, # though he hasn't tried it.
→ Henk has experience with the tofu.

Using question-answer pairs, (51) demonstrates that both with *find* and *like* only the opinion part is the main point of an utterance. Content that can answer explicit or implicit questions is considered QUD at-issue, while content that does not address the QUD is considered QUD not-at-issue (Simons et al. 2010; Tonhauser 2012; Tonhauser et al. 2013). Based on this diagnostic, the experience requirement of *find* (51a) and *like* (51b) is not-at-issue.

- (51) Question 1 (opinion): What do you think about the mapo tofu I made last night?
Question 2 (experience): I made mapo tofu yesterday. Have you tried it?
- a. I found it authentic/delicious.
Question 1:✓, Question 2:#
- b. I liked/loved/couldn't stand it.
Question 1:✓, Question 2:#

To recapitulate, we argue that *find* semantically gives rise to a firsthand experience requirement that is independent of, and is not reducible to, similar requirements imposed by the predicates in its complement (pace Bylinina 2017; Kennedy & Willer 2022; Muñoz 2019). More generally, non-experiential accounts of *find* make incorrect predictions. First, they cannot explain the projection pattern of the AI triggered by *delicious* in the scope of *find*. Second, they do not explain why *find* requires experience even in those cases when the predicate underneath does not, as is the case for *authentic*, *bumpy*, or *challenging*. Finally, these accounts do not predict the infelicity of *find* in stereotypical reasoning cases even though the finder should have enough grounds for their opinion.

Following the ideas expressed by Borkin (1973); Korotkova & Anand (2021); Solt (2018); Stephenson (2007b), we take such data as evidence for experiential *find*. The only formally explicit account in Korotkova & Anand (2021) treats the acquaintance inference of *find* as an instance of direct evidentiary grounding for the complement proposition. In contrast to that, in §4.2 we develop a proposal such that the acquaintance inference of *find* is of the same nature as the experience requirement of psychological predicates, capitalizing on the similarities between *find* and, e.g., *like* discussed in this section. Our point of departure is the idea that both types of predicates express opinion based on experience, with the former being asserted and the latter being backgrounded. More specifically, we formalize the acquaintance inference as the event of experience that (i) connects the finder and the small clause subject and that (ii) causes a change

of opinion in the finder. In §4.3, we provide an additional argument in favor of experiential *find*: *find* is licensed with simple present only when the experience is repeatable. We show that these data fall out naturally under our account, but are difficult to explain under non-experiential approaches.

Defining what exactly counts as experience is admittedly difficult, since it depends on the type of object experienced—abstract objects cannot be experienced through the senses, and yet they can be the embedded subjects of *find*. But they can also be the objects of psychological predicates like *love* and *like* (*I love infinity*), and have similar restrictions on their uses with those. In §5.1, we discuss the obligatory autocentricity of *find* and speculate that it can be derived from experience: *find*, but not other attitude predicates, forces an autocentric reading of subjective predicates in its scope such that the finder is both the opinion holder and the experienter. Finally, §5.2 goes over cases where *find* is licensed seemingly without experience—we argue that those instantiate vicarious experience facilitated by vividness.

3.6. Recap

Let’s take stock. This section provided a detailed overview of the varied empirical properties of English *find*, many of which remained overlooked in the existing literature. Our primary goal was to show that the linguistic behavior of *find* differs from that of *think* in several important aspects that are consequential for how we construe its meaning. Our overarching goal is to demonstrate that several properties of *find*, which at first blush may look peculiar and unrelated, are in fact shared with other verb classes and that looking at *find* through the prism of verbal semantics provides valuable insights into its proper analysis.

- A: small clauses and subject-predicate asymmetries: *find* belongs to the class of characterizing verbs that take small clause complements and exhibit Sæbø’s Puzzle (first pointed out for *find*, but in fact a property of the whole class). We propose that all characterizing small clauses are structured propositions and hypothesize that they have distinct information-structural properties, reflected in the interpretational constraints on subjects of such clauses.
- B: referential transparency and factivity: *find* behaves like emotive factives. It only asserts the subjective/affective predicate in the main predicate position of the embedded clause, while objective content is true both at the world of evaluation and the finder’s belief worlds. The subject position of the embedded clause is only interpreted with respect to higher worlds and is not part of the finder’s mental representation. Only some of these data have been discussed in the literature, and the pattern is not fully predicted by the existing accounts.
- C: lexical aspect: *find* is an achievement predicate, a fact not discussed in the previous literature. We will exploit its event structure to explain its presuppositions and its stative-like interpretation with simple present.
- D: the acquaintance requirement: *find*, like *like* or *hate*, presupposes an experience with an entity that in turn gives rise to the judgment.

In the next section, we provide a unified account of those properties that takes into account the nature of the eventuality described by *find*.

4 Analysis

In what follows, we will provide our semantics in four stages. We will first consider Sæbø’s Puzzle. The cornerstone of our analysis here will be the idea that characterizing verbs, including *find*, select for a structured proposition. This structure will allow us to precisely formulate

the restriction on small clause predicates, which we will link to semantic selection. We will then turn to the aspectual semantics of *find*, which will connect an experiential event to the finding event through causality. We will then consider two applications of this semantics: stative uses of *find* and the complex patterns of ‘de re’ content discussed in §3.2 and §3.3.

Our goal in this section is to delineate several conceptual issues—the segmentation between subject and predicate content, the importance of *find*’s event structure, and the referential transparency and factivity distinctions that *find* makes. In many cases, our concrete semantic decisions are exemplary, and should stand in for a class of similar proposals.

We will assume that expressions in our object language are evaluated relative to structures of the form in (52):

- (52) $M = \langle D_e, D_s, D_\varepsilon, D_\tau, D_T, I \rangle$
 D_e : domain of individuals
 D_s : domain of worlds
 D_ε : domain of events
 D_τ : domain of times
 $D_T = \{0, 1\}$

In addition, we will make use of the following additional product types:

- (53) a. indices: D_i , equivalent to $D_e \times D_s \times D_\tau$
b. propositions: D_{it}
c. event properties: $D_{i\varepsilon t}$

We will assume that propositions are functions from indices to truth values, where indices are $\langle \text{individual, world, time} \rangle$ triples. However, in what follows, we will suppress tense and temporal relations to the extent possible, as well as grammatical aspect. (54) presents the resulting simplified logical form for (23a), repeated below as (54a):

- (54) a. Pascal thought that the documentary was depressing.
b. [Pascal think [the documentary (be) depressing]]

We assume, following [Hacquard \(2010\)](#), that attitude predicates are predicates of events and that every attitude event has corresponding propositional content, retrievable via the CON function. Here, that returns a set of indices compatible with the attitude. This yields the denotation for *think* in (55). Exp (Experiencer) and Theme are semantic roles assigned by predicates to their arguments.

- (55) $\llbracket \text{think} \rrbracket = \lambda p_{it} \lambda x_e \lambda e_\varepsilon \lambda i. \exists e [\text{think}(e, i) \wedge \text{Exp}(x, e, i) \wedge \forall i' \in \text{CON}(e, i) [p(i')]]$

(54b) receives the denotation in (56a), with the embedded proposition denotation in (56b) (where *the documentary* is interpreted ‘de dicto’ by being evaluated relative to the Pascal doxastic indices i').

- (56) a. $\llbracket \text{[Pascal think [the documentary (be) depressing]]} \rrbracket$
 $= \lambda i. \exists e [\text{think}(e, i) \wedge \text{Exp}(\mathbf{P}, e, i) \wedge$
 $\quad \forall i' \in \text{CON}(e, i) [\exists s [\text{depressing}(s, i') \wedge \text{Theme}(\iota z. \text{documentary}(z, i'), s, i')]]]$
b. $\llbracket \text{[the documentary (be) depressing]} \rrbracket$
 $= \lambda i'. \exists s [\text{depressing}(s, i') \wedge \text{Theme}(\iota z. \text{documentary}(z, i'), s, i')]$

4.1. Structured propositions and Sæbø's Puzzle

We begin our proposal with a discussion of the denotations of the small clause complements of characterizing verbs. Recall from §3.2 that we argued that the interpretation of these clauses should separate out the semantics of the subject and predicate. Accordingly, we will assume that these small clauses denote structured propositions of type D_χ , equivalent to $D_e \times D_{iet}$, made up of a subject and an individual property (following Ladusaw 1994 on categorical judgment and Bittner 1998a,b on the complements of *consider*). This gives us the logical form in (57a) for (24c). The small clause translates as the ordered pair in (57b):³⁵

- (57) a. Pascal found [*Moonage Daydream* depressing]
 b. $\llbracket \text{Moonage Daydream depressing} \rrbracket = \langle \llbracket \text{Moonage Daydream} \rrbracket, \llbracket \text{depressing} \rrbracket \rangle$
 $= \langle \mathbf{M}, \lambda i' \lambda x. \exists s [\text{depressing}(s, i') \wedge \text{Theme}(x, s, i')] \rangle$

The split of propositional complement into a subject and property provides us the expressive capacity to capture Sæbø's Puzzle. This asymmetry is challenging for approaches that assess the subjectivity requirement for the entire embedded clause (Stephenson 2007a; Bouchard 2012; Coppock 2018; Kennedy & Willer 2022) and Sæbø uses it as a wholesale assault on the idea that *find* embeds proposition-sized units. The formal challenge posed by Sæbø's Puzzle is that if the attitude embeds a proposition, there is no obvious way to privilege content in a particular part of the embedded clause. However, by splitting the embedded clause into two parts, we are able to target the content coming from the small clause predicate, excluding content from the subject position.

We propose that the asymmetry in Sæbø's Puzzle is best understood in terms of semantic selection, the fine-grained semantic requirements that heads place on their complements (Grimshaw 1979). This includes not only *find*'s subjectivity requirement and prominent recent cases like question-embedding, but also more familiar cases, like prepositions demanding spatial relationship (*onto* requires its object to be a surface) or verbs placing specific requirements on their arguments (*pour* takes liquid-like objects). The best approach to capture semantic selection has been long debated (Asher 2011, 2014; Grimshaw 1979; Katz & Fodor 1963; Pustejovsky 1995), so here we simply state that *find*'s requirements bear family resemblance to many other well-discussed cases and show that this framing allows us to understand its behavior better.

Within the subjectivity literature, the selectional restriction of *find* has frequently been cashed out in terms of whether the property's truth value can vary depending on whatever tracks subjectivity (judges, outlooks, counterstances, etc.). As Sæbø notes, these accounts not only do not distinguish whether subjective content is in the subject or predicate position, but also predict that conjunction/disjunction of subjective and non-subjective predicates should meet *find*'s restrictions, contrary to fact (58a). However, semantic selection by *pour* of its object exhibits the same pattern (58b).

- (58) a. #Pascal found *Moonage Daydream* depressing and fictional.
 b. #I poured the water and the book on the ground.

The bottom line of this discussion is as follows. Once we isolate the predicate from the subject, we can appeal to independent principles of semantic selection to explain the rest of Sæbø's patterns. Accordingly, as stated in the beginning of the paper, we do not take a stand on the specific view on subjective language. Our proposal is compatible with various theories,

35. We take no position on the compositional source for the structured proposition denotation and whether it can be derived from the special syntax of characterizing small clauses (e.g., the one proposed by Basilico 2003).

provided they are supplemented with procedures to account for the facts in (58), and can be implemented using type-theoretic distinctions (Sæbø 2009), assessment-sensitivity (MacFarlane 2014), expressivism about taste (Briesen 2024; Ninan 2024), discretionarity with respect to outlooks (Coppock 2018), or radical counterstance-contingency (Kennedy & Willer 2022). In what follows, we will simply gloss this restriction as a requirement that the individual property is SUBJECTIVE, as in the skeletal semantics in (59). Since the small clause is a structured proposition, it is possible to state the requirement of subjectivity as holding only over the individual property, which we add for concreteness as a presupposition.

(59) $\llbracket \text{find} \rrbracket = \lambda \langle S, P \rangle \in D_\chi \lambda x \lambda e \lambda i :$

- a. presupposes:
SUBJECTIVE(P)
...
- b. asserts:
....

What remains is to fill in those dots, and then to understand how that can help us derive the lack of ‘de dicto’ material in *find*.

4.2. The semantics of *find*: aspect and experience

In §3, we argued that *find* is a simple achievement that presupposes an event where the finder experiences the embedded subject and asserts the subjective property expressed by the material in the main predicate position.

Let us start with the idea that *find* is an achievement, which we will treat as a change of state. We will use the proposal in Beavers & Koontz-Garboden (2020), which situates Dowty’s (1979) compositional treatment of word meaning within event semantics. In particular, we will make use of two primitives, BECOME and CAUSE. The semantics for BECOME is provided below:

(60) BECOME(s, e, i) is True iff at the beginning of e the state s does not hold and at the end of e the state s does hold. (Beavers & Koontz-Garboden 2020:36)

BECOME as a property of an event and a state requires that the state did not hold prior to the event, and that it holds after. As McNally (2024) notes, this requires the becoming event e to be like what Pustejovsky (1995) calls a *transition*, an immediate sequencing of two contrary states. Along with Pustejovsky, we take transitions to ontologically be non-durative, which explains why they are not compatible with durative adverbials or the progressive.

With that in mind, we can provisionally analyze *find* as in (61). Though we return to it in §4.4, note that the expression $P(i')(S)$ already prevents the subject from being read ‘de dicto’.

(61) $\llbracket \text{find} \rrbracket = \lambda \langle S, P \rangle \in D_\chi \lambda x \lambda e \lambda i :$

- a. presupposes:
SUBJECTIVE(P)
- b. asserts:
 $\exists s [\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \forall i' \in \text{CON}(s, i) [P(i')(S)]]$

(61) states that the event e is a transition to a novel state s . This new state has propositional content, and its content meets the structured proposition the small clause denotes: the indices in the content of s are ones which make the small clause individual property true, when evaluated

at the small clause subject.

This amounts to saying that events of finding are transitions into new attitudinal states, which are of category FS (find state). We do not probe what precise type of cognitive state these are.

We now turn to the experiential requirement. As discussed in §3.5, this requirement is a presupposition. Accordingly, we will assume a presupposed event of experience, e_{exp} , with the finder as experiencer and the small clause subject is the theme, as in (62):

- (62) $\llbracket \text{find} \rrbracket = \lambda \langle S, P \rangle \in D_\chi \lambda x \lambda e \lambda i :$
- a. presupposes:
 $\text{SUBJECTIVE}(P) \wedge$
 $\exists e_{exp} [\text{Exp}(x, e_{exp}, i) \wedge \text{Theme}(S, e_{exp}, i)]$
 - b. asserts:
 $\exists s [\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \forall i' \in \text{CON}(s, i) [P(i')(S)]]$

Note that the representation in (62) does not link the experiential event to the change of state event that *find* describes. We have argued that *find* encodes an evidential link between e_{exp} and e —the transition to the corresponding state happens because of e_{exp} : e_{exp} provides some content (perhaps non-propositional) which leads to the corresponding characterization. While we believe there is a richer description of the processes that describe this cognitive pathway (see also §5.1), for our present purposes it suffices to say that e_{exp} *causes* e to occur, and so we will make use of the primitive relation (directly) CAUSE between events c (*cause*) and e (*effect*):

- (63) $\text{CAUSE}(c, e, i)$ is true iff c directly causes e in i .

Fully fleshing out the causal structure of this pathway is beyond the scope of this paper (see Bar-Asher Siegal 2026 for discussion), but we will assume that CAUSE is a species direct causation (Fodor 1970b; Shibatani 1976), which requires the link between the c and e to be unmediated by intermediate causes (for critical discussion on this idea, see Baglini & Siegal 2025; Neeleman & van de Koot 2012; Wolff 2003).

We can thus add to the presuppositional component of *find* that e_{exp} causes e . *Find* makes additional presuppositional demands of e as well. Like *think*, *find* is a neg-raising verb. Negation of x *find* Y leads not simply to denial of x having an change of state to a find state with content Y . It moreover requires that x end up in a find state with content $\neg Y$. Thus, *Pascal didn't find Moonage Daydream depressing* is equivalent to *Pascal find Moonage Daydream not depressing*. Following Gajewski (2007) and Bartsch (1973), we will assume that neg-raising is the product of a presupposition of SETTLEDness (or opinionatedness) of the attitude holder's information state (64). This excludes the possibility that the information state is undetermined on the issue in question. In other words, a *find*-report always means that the finder is decided about the complement.

- (64) For propositions q and p ,
 $\text{SETTLED}(q, p)$ iff $q \subseteq p \vee q \subset \neg p$.

In (65), we state that the content of the find state s is SETTLED on the small clause content. Thus, e is a transition to state s which resolves the issue of whether the property holds of the subject:

- (65) $\llbracket \text{find} \rrbracket = \lambda \langle S, P \rangle \in D_\chi \lambda x \lambda e \lambda i :$
- a. presupposes:

- SUBJECTIVE(P) $\wedge$
 $\exists e_{exp}, s[$
 $\text{Exp}(x, e_{exp}, i) \wedge \text{Theme}(S, e_{exp}, i) \wedge \text{CAUSE}(e_{exp}, e, i) \wedge \text{BECOME}(s, e, i) \wedge$
 $\text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \text{SETTLED}(\text{CON}(s, i), \lambda i'. P(i')(S))]$
b. asserts:
 $\exists s[\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \forall i' \in \text{CON}(s, i)[P(i')(S)]]$

Armed with this, we can complete the derivation for (57):

- (66) a. Pascal found [*Moonage Daydream* depressing]
b. $\llbracket [\text{Moonage Daydream depressing}] \rrbracket = \langle \mathbf{M}, \lambda i' \lambda x. \exists s[\text{depressing}(s, i') \wedge \text{Theme}(x, s, i')] \rangle$
c. $\llbracket \text{Pascal find [Moonage Daydream depressing]} \rrbracket =$
 $\llbracket \lambda \langle S, P \rangle \in D_X \lambda x \lambda e \lambda i :$
SUBJECTIVE(P) $\wedge$
 $\exists e_{exp}, s[$
 $\text{Exp}(x, e_{exp}, i) \wedge \text{Theme}(S, e_{exp}, i) \wedge \text{CAUSE}(e_{exp}, e, i) \wedge \text{BECOME}(s, e, i) \wedge$
 $\text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \text{SETTLED}(\text{CON}(s, i), \lambda i'. P(i')(S))]$
 $\exists s[\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \forall i' \in \text{CON}(s, i)[P(i')(S)]]]$
 $\langle \langle \mathbf{M}, \lambda i' \lambda x. \exists s[\text{depressing}(s, i') \wedge \text{Theme}(x, s, i')] \rangle \rangle$
 $= \lambda e \lambda i :$
SUBJECTIVE(P) $\wedge$
 $\exists e_{exp}, s[$
 $\text{Exp}(\mathbf{P}, e_{exp}, i) \wedge \text{Theme}(\mathbf{M}, e_{exp}, i) \wedge \text{CAUSE}(e_{exp}, e, i) \wedge \text{BECOME}(s, e, i) \wedge$
 $\text{FS}(s, i) \wedge \text{Exp}(\mathbf{P}, s, i) \wedge$
 $\text{SETTLED}(\text{CON}(s, i), \lambda i'. \exists s[\text{depressing}(s, i') \wedge \text{Theme}(\mathbf{M}, s, i')])]$
 $\exists s[\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(\mathbf{P}, s, i) \wedge$
 $\forall i' \in \text{CON}(s, i)[\exists s'[\text{depressing}(s', i') \wedge \text{Theme}(\mathbf{M}, s', i')]]]$.

We thus suggest that *find* presupposes not just e_{exp} , but also a causal structure starting from e_{exp} which results in the onset of a find state s in which either the small clause holds or fails to hold. And then what is asserted is that it holds. In tying e_{exp} causally to the small clause, the semantics imposes restrictions on e_{exp} itself: it must be sufficient to ground the small clause content for the finder via direct causality.

We argued in §3.5 that while *find* lacks a narrow experience requirement (since it is, e.g., acceptable in cases where there is no sensory evidence, like (48)), it nonetheless requires the experience to be sufficient to causally ground the resulting state, and this helps to constrain the otherwise unspecified experience. For example, one cannot find a cup of tea delicious by looking at it, even if it's possible to reliably infer from stereotypical reasoning that it will be delicious. We thus suggest that the direct causal relationship indicated by CAUSE in the semantics of *find* forecloses mediation by inferential reasoning (which we view as an intermediate causal state). Similarly, one cannot find a picture scary by description, since that description does not qualify as an experience of the picture itself. But one can find an academic boycott meaningless via description because the description is both an experience of the (idea of the) boycott and sufficient to causally ground the resulting affect. In this we agree with Kennedy & Willer (2022) that it is too constraining to specify what counts as experience, since it has to be modulated to the property in question (see also Cariani 2022 on ideal grounding). For many predicates, the causal conditions are very constrained. But for those like *challenging* and *authentic*, which lack any lexical acquaintance conditions, what counts as an experience is actually quite broad. One can find a street bumpy by riding over it or tactile encounter, but

potentially also by looking at a detailed image or video of people encountering it.

The problem of specifying what counts as a sufficient kind of experience is thus highly complex, and a proper discussion will take us too far afield here. We note, however, that this same complexity is shared by psych predicates as well. As mentioned in §3.5, [Cohen & Erteschik-Shir \(2002\)](#) argue that psychological predicates require acquaintance. For example, *John hates lawyers* presupposes that John is acquainted with lawyers. But what counts as acquaintance here? Just knowing that they exist? Or what they are? Or encountering lawyers? Or their effects on the world? All of these kinds of acquaintance are mirrored with different small clause properties of *find*. Thus, *John finds lawyers annoying* vs. *John finds lawyers charming* bring to mind different kinds of encounters with lawyers.

In sum, our semantics for *find* links together an experiential encounter between the finder and the small clause subject, the caused change of state that *find* denotes, and the content of the resultant state. The aspectual status of *find* is thus a clue to the underlying event structure, and the causal semantics provides a basis for the otherwise puzzling grounding restrictions *find* imposes.

In the following two sections we will explore how this semantics interacts with habituality to yield stative interpretations of *find* and with ‘de re’ interpretation to yield the transparency and factivity inferences we discussed in §3.

4.3. The stative *find* conspiracy

In §3.4, we have shown that *find* is an achievement, which we have captured with change of state semantics in §4.2. At the same time, *find* often masquerades as a state with the simple present: nearly synonymous to *think*, it seems to express a general opinion of the finder about an entity, such as [Sæbø’s \(2009\)](#) *She finds all smokers unpleasant*. Many examples in the literature are precisely of this sort, which creates an impression that *find* is more similar to *think* than it actually is. We argue that such stative-like readings are intrinsically linked *find*’s experiential semantics (§3.5): these readings can arise only when the experience that causally grounds the change of state (i) is circumstantially repeatable and (ii) actually took place. Based on this, we propose that stative uses of *find* are habituals, licensed by a covert stativizing habitual operator.

We begin with repeatability. Consider a case where the finder can no longer experience—or thinks that they no longer can experience—the entity under evaluation in the relevant way for the property in question.³⁶ In (67) (modeled after the case of Jan Rodrick in Arthur C. Clarke’s *Childhood’s End*), while the Earth exists, the finder can no longer behold it, or at least thinks that they can no longer behold it. The present tense is felicitous in the unembedded case and with *consider* (67c). However, the present tense is impossible with *find* (67a).

36. *Find* gives rise to the so-called lifetime effects ([Musan 1997b](#)), requiring the entity in question to exist at the reference time. These effects follow from two factors: (i) the predicates that *find* selects for, often classified as individual-level (cf. footnote 7) and known to give rise to lifetime effects when unembedded ([Carlson 1980](#)), and (ii) the syntax of small clauses, which lack their own tense. When used in present tense, *find* suggests that the entity under evaluation still exists. *Find* is therefore unacceptable with individuals that no longer do, such as a shooting star that the finder saw the previous day (1).

(i) *We’re discussing last night’s meteor shower and especially one unusually bright purple star.*

a. I #find/✓found the purple shooting star glorious/ominous.
b. The purple shooting star #is/✓was glorious/ominous.

As the example above shows, *find*’s behavior mirrors the pattern with unembedded predicates like *glorious*. (67) is different. Such examples ensure that the embedded subject entity is in existence at the reference time for *find*, which in turn demonstrates that the infelicity is not due to the lifetime effect.

- (67) *A stowaway on an alien ship, leaving Earth for good, muses about what they'll miss. They couldn't bring any photographs, and there is no technology that would allow them to see their home planet.*³⁷
- a. #I find the Matterhorn beautiful/easy to climb, even though I'll never see it again.
 - b. ✓The Matterhorn beautiful/easy to climb, even though I'll never see it again.
 - c. ✓I consider the Matterhorn beautiful/easy to climb, even though I'll never see it again.

We suggest that the pattern is due to the conflict between the experience requirement of *find* and the habitual interpretation of the English simple present with non-states: the habitual requires iterativity, *find* requires a causing experiential event, and together they require that experience for each iteration of finding. Thus, acceptable uses of *find* in the simple present must be habituals. A habitual predication requires that the predication is circumstantially possible in the the future. In (67a), because the stowaway is not able to return to Earth, they cannot re-experience the Matterhorn, and thus cannot use *find* to describe their subjective state.

In thinking about this property of habituals, we follow Ferreira (2016) and assume that habituality is provided by a silent *HAB* morpheme in English. Ferreira argues that habituals require instantiation of a plurality of events that meet the verb phrase description ($PL(\mathcal{P}(w'))(e)$), holding over an interval that contains the reference time; it is that extended plurality that licenses the use of the simple present. And this extended plurality influences the truth value of habitual sentences. In (68) below, the habitual *John plays soccer* is felicitous if John has the circumstantial ability to play soccer repeatedly in the future (Context 1). However, if that ability is obstructed by the field closing (Context 2) or a change in John's own abilities (Context 3), the habitual is no longer felicitous.

(68) The Soccer player

Context 1 (typical): *John lives close to the only soccer field in the neighborhood, and regularly plays there.*

Context 2 (the closed field): *While John loves soccer, the only field in his neighborhood closed years ago and he does not play regularly.*

Context 3 (the tragic accident): *While John loves soccer, he has suffered an accident and can no longer control his legs.*

John plays soccer (regularly).

Context 1: ✓, Context 2: #, Context 3: #

Note that habituals are a species of modals that are not foreclosed about the future in general (for example, one can truthfully utter *John plays soccer* even if it is known that he will die eventually, as all humans do). Building on Portner's (1998) semantics for the progressive, Ferreira proposes that the habitual is a modal involving quantification of a circumstantial modal base M characterizing John's current physical/mental conditions, the local circumstances, and his past soccer-playing, and that this is ordered by O , 'normal' non-interruption conditions (e.g., no injuries, calamities, etc.) BEST holds of the closest worlds in modal base M , ordered by ordering source O and the requirements of the predicate:

- (69) $\lambda w. \forall w' \in \text{BEST}(\mathcal{P}, M, O, w, t) [\exists e_{\oplus} : PL(\mathcal{P}(w'))(e_{\oplus}) \wedge t* \subseteq \tau(e_{\oplus})]$, where
 $\mathcal{P} = \lambda w. \lambda e. e$ is an event of John playing soccer in w . (Ferreira 2016)

In Context 1, $M_1 = \{\text{John played soccer several times recently, John is in good physical$

37. The Matterhorn is a mountain in the Swiss Alps.

condition, There is a field near John's house, John intends to play soccer again, ...} and $O_1 = \{\text{John does not die tomorrow, John does not incur injuries, The stadium does not close, ...}\}$. The semantics in (69) then demands of the closest M worlds ordered by O that there is a plurality of events e where John plays soccer, where e 's temporal trace contains the utterance time. This requires that in those worlds e is instantiated both before and after the utterance time, which is possible given M_1 and O_1 . And because of O_1 , we consider ideal circumstances, ignoring any possible changes to John or the field. In Contexts 2 and 3, however, because those are current facts, they enter into M . In Context 2, M_2 includes the fact that the field is closed and in M_3 worlds are ones where John's accident occurs. In these worlds, there is no guarantee that John plays soccer in the future, however ideal the ordering is. As a result, the semantics in (69) is not met, blocking the truth of the habitual assertion.³⁸

We propose that a similar story explains the unacceptability of (67). We have seen that a present habitual requires that the worlds meeting the circumstantial restrictions in M are ones where the stowaway can find the Matterhorn beautiful in the future. But now, if *find* is an achievement and describes a change of state, that change must be possible in the future, and it must be caused by a novel experience of the Matterhorn, which must occur after the utterance time. Thus, in order to use (67a), the stowaway must be capable of experiencing the Matterhorn in the future, something they are circumstantially precluded from by their one-way ticket on an alien ship (parallel to Context 2 in (68)). Thus, we correctly predict the infelicity of (67a) due to the fact that a present habitual cannot be realized in the future. We provide the semantics for habitual *find* in (70); for readability, we have suppressed discussion of find state s and SETTLEDness in the presuppositional content.

$$(70) \quad \llbracket HAB [\text{Jan find} [\text{Matterhorn beautiful}]] \rrbracket = \\ \lambda t \lambda i. \forall i' \in BEST(\mathcal{P}, M, O, i) [\exists e_{\oplus} : PL(\mathcal{P}(i'))(e_{\oplus}) \wedge t \subseteq \tau(e_{\oplus})], \text{ where} \\ \mathcal{P} = \lambda i. \lambda e : \\ \text{SUBJECTIVE}(\lambda i \lambda x. \exists s [\text{beautiful}(s, i) \wedge \text{Theme}(\mathbf{M}, s, i)]) \wedge \\ \exists e_{exp} [\text{Exp}(\mathbf{J}, e_{exp}, i) \wedge \text{Theme}(\mathbf{M}, e_{exp}, i) \wedge \text{CAUSE}(e_{exp}, e, i)] \\ \cdot \exists s [\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(\mathbf{J}, s, i) \wedge \forall i' \in \text{CON}(s, i) [\\ \exists s' [\text{beautiful}(s', i') \wedge \text{Theme}(\mathbf{M}, s', i')]]]]$$

Our account of this infelicity is crucially dependent on the idea that *find* describes a change of state that is motivated by experience. We thus predict that expressions which lack that experiential requirement should be acceptable in the stowaway scenario. As we see in (67b), *consider*, which we have shown does not come with an experience requirement, is acceptable. Furthermore, experiential expressions that are not achievements are predicted to be felicitous.

38. The literature on genericity distinguishes two potential readings of simple present forms with non-states (see Boneh 2019 and Cable 2022 for recent discussions): (i) habituals, which describe an ongoing series of events that meet the description, and (ii) dispositionals (also referred to as ‘characterizing sentences’), which describe a property that may be instantiated. One of the hallmarks of dispositionals is that they do not require verifying instances (as with Krifka et al.’s (1995) famous dispositional *Mary sorts the mail from Antarctica*). Note that *find* sentences never seem to allow a dispositional (verifying instance-free) reading: (1) is not acceptable if Jan has never encountered mountains before.

(i) #Jan finds mountains beautiful.

Menéndez-Benito (2013) observes that dispositional, but not habitual, readings can often be paraphrased with circumstantial modals. It is thus relevant that circumstantials are not very acceptable with *find*, which further supports our proposal that present tense *find* is habitual:

(ii) #Pascal can find mountains/the Matterhorn beautiful.

This prediction is borne out. While psychological predicates like *like* also presuppose experience (§3.5), their aspectual properties are different.

(71) I like the Matterhorn even though I will never see it again.

As an achievement, *find* describes a change of state caused by an experiential event. It is because we need to instantiate a new change of state, and hence need a new event of experience, that we require the stowaway to re-experience the Matterhorn, leading to the infelicity of (67a). In contrast, stative psych predicates such as *like* can be true in the simple present without requiring iteration of some dynamic process. Hence, one can like something without being obliged to commit to the circumstantial possibility of re-experience.

Let us take stock. We have shown that present tense *find*, while very common overall, is constrained in a way neither *think* nor *consider* are. Because *find* is an achievement that lexically encodes experience, it is felicitous with simple present only when experience is potentially repeatable due to the plurality requirement of habituals. When this condition is not met (for example, when the entity no longer exists or when the finder believes they may no longer access it), present-tense *find* is infelicitous due to a presupposition failure.

Any account of *find* that includes, or is compatible with, a lexical experiential component (whether asserted or presupposed) will get our data (e.g., Bouchard 2012; Coppock 2018; Korotkova & Anand 2021; Stephenson 2007b). At the same time, the pattern discussed in this section presents an additional problem for those approaches that explicitly argue against experiential *find* (we discuss other issues with those views in §3.5). Thus, those approaches that treat *find* as selecting for experiential predicates predict that *find* will not be different from unembedded cases, contrary to fact: while (67b) is good, (67a) is bad. Likewise, Kennedy & Willer (2022) argue that the experience requirement with *find* arises from the combination of the radical counterstance-contingency of *find* and the affective grounding of the predicate in the complement clause. Their account does not directly incorporate lexical aspect, however, even if it is modified in such a way that *find* is an achievement, this fact alone does not explain why present-tense *find* is infelicitous in cases like (67a). There is nothing in the stowaway scenario that, in principle, prevents the finder from a habit of habitually revising their judgment (e.g., falling in love with the Matterhorn, over and over). The problem is that the finder cannot do so if doing so requires re-experiencing the mountain. While it may be possible to marry the account in (Kennedy & Willer 2022) with our empirical observations without committing an experiential *find*, the account makes incorrect predictions in its present state. We treat such data as yet another argument in favor of our preferred analysis that treats the event structure of *find* seriously.

4.4. Non-subjective content: making the right cut

We now turn to the question of non-subjective content in the scope of *find*. Recall what we observed in §3.2 and §3.3. We noticed that the indices of evaluation for content in the scope of *find* depend on two factors: the position of the content (subject vs. predicate position) and the nature of the content (subjective vs. non-subjective). In subject position, all content is mandatorily interpreted with a ‘de re’ construal, and is evaluated relative to the matrix index of evaluation. In predicate position, there is a split. Non-subjective content is interpreted relative to *both* the index of evaluation and the finder. In contrast, subjective content in predicate position is only interpreted relative to the finder. When we add the factor of presupposed vs. asserted content, there are three distinct patterns we aim to capture. The first is that *find* is factive with respect to the objective content of the small clause property, as noted before (Bouchard 2012; Cop-

pock 2018). Additionally, recall, as with emotive factives, it is also presupposed that the finder is themselves committed to the objective content of the small clause property as illustrated in (33-34) and (36) in §3.3. We repeat (33) below:

(33) Pascal found *Moonage Daydream* an uplifting documentary.

- a. ...#even though it is a mockumentary.
→ *Moonage Daydream* is a documentary in w_0 .
- b. ...#even though he thinks it is a mockumentary.
→ Pascal thinks that *Moonage Daydream* is a documentary.
- c. ...# even though he also found it depressing.
→ Pascal judged the movie uplifting.

In some sense this is a reasonable property: given that the aim of subjective *find* is to characterize the subject with respect to a subjective property, it would seem reasonable for the finder to be aware of properties that are relevant for that characterization. How can one construe a film as uplifting except by referencing its comparison class?

At the same time, though, it does seem intuitively possible to judge whether a film is uplifting relative to films, without thinking of it as a documentary. The crucial point that (33) shows us is that when the nominal is part of the small clause property, it both becomes part of the finder's criteria for their judgment and must also hold in the evaluation world. If the objective content is removed from the small clause, neither of these conditions must be present. For example, a modifier clause can explicitly provide the comparison class the finder is using, as in (72a) or state the finder's ignorance (72b).

- (72) a. Thinking *Moonage Daydream* was a mockumentary, Pascal found it uplifting.
↗ *Moonage Daydream* is a mockumentary.
- b. Not knowing that *Moonage Daydream* was a documentary, Pascal found it uplifting.
↗ Pascal believes that *Moonage Daydream* is a documentary.

The cases in (72) stand in contrast to the contradictions we observed in (33) for objective content. And what differs between these cases is that the complement clause in (33) entails that S bears the objective predicate, while the complement in (72) does not.

Thus, for the small clause predicate, we will follow Bouchard (2012) in proposing that the semantics of *find* splits the content, presupposing the truth of the objective content of the small clause property. However, it does so twice, once for the index of evaluation and once for the finder's indices. In order to do this, we modify the truth conditions in (65) minimally so that the presupposed find state s has content which entails that the objective content of P is true for for subject S .³⁹ We appeal to $\mathcal{O}(P)$, which returns the maximal objective entailed property Q of a property P , as defined in (73) below:

$$(73) \quad \mathcal{O}(P) = \max_Q \{Q : \forall i, x [P(i)(x) \rightarrow Q(i)(x)] \wedge \neg \text{SUBJECTIVE}(Q)\}$$

$$(74) \quad \llbracket \text{find} \rrbracket = \lambda \langle S, P \rangle \in D_X \lambda x \lambda e \lambda i :$$

- a. presupposes:
SUBJECTIVE(P) $\wedge$
 $\exists e_{exp}, s [$
Exp(x, e_{exp}, i) $\wedge$ Theme(S, e_{exp}, i) $\wedge$ CAUSE(e_{exp}, e, i) $\wedge$ BECOME(s, e, i) $\wedge$

39. We may alternatively appeal to the finder's doxastic indices. As discussed above, we do not sufficiently understand the difference of these two states to reliably determine which is relevant here.

$$\text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \text{SETTLED}(\text{CON}(s, i), \lambda i'. P(i')(S)) \wedge \\ \mathcal{O}(P)(i)(S) \wedge \forall i' \in \text{CON}(s, i) [\mathcal{O}(P)(i')(S)]$$

b. asserts:

$$\exists s [\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge \forall i' \in \text{CON}(s, i) [P(i')(S)]]$$

The semantic content of S is dealt with differently, since its content is evaluated solely in the evaluation index i (as [Sæbø 2009](#) demonstrates). This is indicated in semantics in (74) by the fact that $\mathcal{O}$ does not apply to S .

With this semantics for *find*, we proceed to derive the truth conditions for (75).

(75) Pascal found the movie Mordecai recommended an uplifting mockumentary.

The small clause in (75) has the denotation in (76b).⁴⁰ We abbreviate its subject by (76c) in the formulas to follow, and its property by (76d):

- (76) a. [the movie Mordecai recommended] an uplifting mockumentary]
 b. $\llbracket$ [the movie Mordecai recommended] an uplifting mockumentary $\rrbracket =$
 $\langle \iota z [\text{movie}(z, i) \wedge \exists e [\text{recommend}(e, i) \wedge \text{Agent}(\mathbf{M}, e, i) \wedge \text{Theme}(z, e, i)]]],$
 $\lambda i' \lambda x. \exists s [\text{uplifting}(s, i') \wedge \text{Theme}(x, s, i') \wedge \text{mockumentary}(x, i')] \rangle$
 c. $\llbracket$ the movie Mordecai recommended $\rrbracket = \iota z [\text{rec-movie}(z, \mathbf{M}, i)]$
 d. $\llbracket$ an uplifting mockumentary $\rrbracket = \lambda i' \lambda x. \text{uplifting-mock}(x, i')$

Correspondingly, $\mathcal{O}(P) = \lambda i' \lambda x. \text{mockumentary}(x, i')$. With this in place, we can examine the whole sentence:

- (77) a. Pascal find [[the movie Mordecai recommended] an uplifting mockumentary].
 b. $\llbracket$ Pascal find [[the movie Mordecai recommended] an uplifting mockumentary $\rrbracket =$
 $\lambda e \lambda i :$
 $\text{SUBJECTIVE}(\lambda i' \lambda x. \text{uplifting-mock}(x, i')) \wedge$
 $\exists e_{exp}, s [$
 $\text{Exp}(x, e_{exp}, i) \wedge \text{Theme}(\iota z [\text{rec-movie}(z, \mathbf{M}, i)], e_{exp}, i) \wedge \text{CAUSE}(e_{exp}, e, i) \wedge$
 $\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge$
 $\text{SETTLED}(\text{CON}(s, i), \lambda i'. \text{uplifting-mock}(\iota z [\text{rec-movie}(z, \mathbf{M}, i)], i')) \wedge$
 $\text{mockumentary}(\iota z [\text{rec-movie}(z, \mathbf{M}, i)], i) \wedge$
 $\forall i' \in \text{CON}(s, i) [\text{mockumentary}(\iota z [\text{rec-movie}(z, \mathbf{M}, i)], i')]]$
 $\exists s [\text{BECOME}(s, e, i) \wedge \text{FS}(s, i) \wedge \text{Exp}(x, s, i) \wedge$
 $\forall i' \in \text{CON}(s, i) [\text{uplifting-mock}(\iota z [\text{rec-movie}(z, \mathbf{M}, i)], i')]]$

This resulting semantics presupposes an experiential event e_{exp} that causes a change of state e , with the resulting state s one that settles whether the movie Mordecai recommended in i is an uplifting movie in the i' content indices associated with s . But it also presupposes that that recommended movie in i is a mockumentary, both in i and in those same i' content indices. As a result, settling the issue ultimately involves settling the subjective component of being an uplifting mockumentary, since those i' are already settled that the movie is a mockumentary. The at-issue content in turn asserts that the movie is an uplifting mockumentary in i' .

Thanks to the structured proposition denotation of the small clause, we thus can derive the pattern we started this section with. The small clause subject is uniformly evaluated exclusively

40. In (76), we assume that the subject is interpreted relative to the evaluation index i . This is a consequence of the factive semantics of *find*; other characterizing verbs allow ‘de dicto’ subjects. Thus, we may assume that S is an individual concept, and that where we write S in (74), we should use $S(i)$.

at i , while the content of the small clause predicate shows a split: the objective content is presupposed to be true at both i and i' , and the subjective content is asserted to be true at i' . The fact that the subject is evaluated only at i ensures the mandatory ‘de re’ interpretation. In the case of (75), it means that Pascal need not know that Mordecai recommended the film.

And when we insert a subjective predicate in the subject, as in (75), the subjective *depressing* is evaluated only in i , while the main predicate *uplifting* is evaluated only with respect to i' worlds.

- (78) Pascal found the depressing movie Mordecai recommended an uplifting mockumentary.
 $\nrightarrow$ Mordecai recommended the movie in Pascal’s i' indices.
 $\nrightarrow$ the movie is depressing in Pascal’s i' indices.
 $\nrightarrow$ the movie is uplifting in i .
 $\rightarrow$ the movie is a mockumentary in i .

But the objective *mockumentary* is presupposed as true in i , which leads to a contradiction in scenarios where the property fails to hold of the subject, or if the subject’s content contains a contrary property (e.g., *documentary*). We thus derive the contradictions we observed in (24):

- (24) a. ✓Pascal thought that the uplifting documentary was a depressing mockumentary.
b. #Pascal found the uplifting documentary a depressing mockumentary.
c. #Pascal found *Moonage Daydream* a depressing mockumentary.
d. #Pascal found *Moonage Daydream* a depressing documentary _{i} .

(24c) requires that the movie is a mockumentary, contrary to fact. The subject content in (24b) requires that the referent is a documentary, but the property content requires that it’s a mockumentary. And (24d) requires that the movie is a documentary in Pascal’s i' indices, contrary to the scenario.

In sum, we derive a semantic distribution for *find* as a hybrid between factive attitudes like *discover* and non-factive attitudes like *think*. *Think* has no requirements about the truth of embedded content with respect to the evaluation index i , which allows us to counterpose an attitudinal state with the evaluation index, as in (24a). And whereas *discover* requires the attitude holder and i to agree on embedded clause content (modulo ‘de re’ interpretation), *find* does so only for $\mathcal{O}(P)$ (§3.3), the objective content of the embedded property.

5 Loose ends

This section goes over several aspects of *find*’s behavior for which we do not offer a full-fledged solution to in this paper, along with tentative directions as to what such solutions might be. We start with causality and then move on to the cases of vicarious experience.

5.1. Causality

We have shown that *find* is an achievement predicate. Achievements are typically understood as transitions from one state to another, and our proposed account uses lexical decomposition in the spirit of Dowty (1979), with the change of state due to the presence of semantic primitive CAUSE in the semantics of *find*. Recent approaches to causality in natural language have demonstrated the advantages of incorporating causal models into the semantics of various causation-sensitive expressions, including change-of-state eventualities and a variety of causative constructions (see overview in Bar-Asher Siegal 2026). While we do not use this framework here for the sake of simplicity of representation, we are sympathetic to the over-

all project and would like to point out two specific features of *find* that can be understood via constraints on causal chains: its factivity and the interpretation of subjective expressions in its scope.

Factivity We have argued that *find* is a factive predicate that gives rise to inferences about truth and belief (§3.2). Our analysis simply stipulates which inferences are not-at-issue. More specifically, the only at-issue, non-presupposed component of *find* is about the finder’s opinion about an entity. All other meaning components are presupposed, including the entity’s existence, the finder’s mental representation of its objective properties and the fact that the subjective judgment was caused by an event of encounter with said entity.

We hypothesize that this factivity pattern can be derived by an approach to presupposition triggering that specifies default principles of content backgrounding which make reference to *find*’s causal network. Under the account of Abrusán (2011), entailed content not about the projected event is by default backgrounded, becoming a presupposition. The key observation for *find* is that all of the presupposed content lies causally upstream of the finder’s change-of-state. If entailments about events causally upstream from that change-of-state are not about the change-of-state, then they will be backgrounded. Such an account thus correctly predicts that only the change-of-state will be at-issue.

This explanation aligns with recent approaches to factive predicates (*discover, know, regret*) and presupposition projection at large that attempt to predict which entailments become not-at-issue without treating them as lexical presuppositions. More specifically, factive predicates have been argued to give rise to an at-issue entailment about preadjacent truth that often is backgrounded due to the interaction with larger discourse context and grammatical environment (see discussion in Scontras & Tonhauser 2026). We leave a detailed exploration of possible connections between the causality of *find* and its projective behavior for future research, but highlight this avenue as promising.

Obligatory autocentricity Consider (79). Unembedded subjective predicates allow a non-autocentric reading such that the expressed judgment is not that of the speaker and may not even be shared by the speaker (Lasersohn 2005), easiest to see when we’re talking about non-human species or other situations when the speaker is irrelevant (Egan et al. 2005; Pearson 2013a; Stephenson 2007b). Thus, the unembedded *tasty* in (79a) is easily construed in a way such that both the experience and the evaluation are not anchored to the speaker.⁴¹ It behaves the same way in attitude reports, as first pointed out Lasersohn (2005) and discussed in detail by Anand & Korotkova (2022): *tasty* is interpreted relative to the local worlds, but it doesn’t have to reflect the attitude holder’s opinion or experience (79b). As first shown by Bouchard (2012), *find* is more rigid: a subjective predicate in the scope of *find* always reflects the finder’s experience and opinion (79c).

(79) *Our pet dragon is a finicky eater. We’re constantly experimenting with various foods to see what he likes.*

- a. This new brand of salted glass is tasty—just look at how he eagerly waits to be fed.
 ↗ The speaker has tried salted glass. [non-autocentric experience]
 ↗ The speaker liked salted glass. [non-autocentric opinion]
- b. Henk is glad/thinks/observed/noticed that this new brand of salted glass is tasty.

41. (79) is modelled after Stephenson’s (2007b) *cat food* examples. We use *salted glass* as a test case to highlight the relevant contrasts, as this is something that humans can’t consume, unlike cat food that they simply don’t consume normally.

- ↗ Henk has tried salted glass. [embedded non-autocentric experience]
- ↗ Henk liked salted glass. [embedded non-autocentric opinion]
- c. Henk finds/found this new brand of salted glass tasty.
 - Henk tried salted glass. [embedded autocentric experience]
 - Henk liked salted glass. [embedded autocentric opinion]

There are several issues raised by the examples in (79). The first one is technical and requires one to take a stand on the proper semantics of subjective expressions. More specifically, we need to get the compositional semantics right so that it correctly predicts the non-autocentricity of *tasty* in matrix cases and embedded cases without overgenerating for *find*. The accounts in Coppock (2018); Lasersohn (2005) overgenerate for *find*, as they do not require an embedded autocentric reading in intensional environments. The accounts in Sæbø (2009); Stephenson (2007b); Stojanovic (2007) may deal with *find*, but they overgenerate for *think* (see detailed discussion in Anand & Korotkova 2022). The account in Pearson (2013b) undergenerates for *think*, as it only allows embedded autocentric readings. The accounts in Franzén (2018, 2021); Khoo (2026) correctly predict the behavior of *tasty* under *find* as they don't derive *find* from *think*, but at the same time they do not discuss not-autocentric matrix uses of *tasty* and thus may have problems in explaining (79a).

However, there is also a more interesting conceptual issue: the connection between the presence of an experience and the judgment it causes. Even though *find* does not behave like classic attitude predicates, it patterns exactly like psychological predicates such as *hate*, *like* or *love*, which also encode opinion based on experience (a fact reflected in their argument structure; see also discussion in §3.5). (80) illustrates that the subject of *like* must be both the experiencer and the opinion holder.

- (80) *Salted glass is something that dragons consume, and Henk noticed that his dragon is particularly excited about this new brand.*
- a. Henk likes this new brand of salted glass #though he hasn't tried it himself.
 - Henk experienced this brand of salted glass. [subject-centric experience]
 - b. Henk likes this new brand of salted glass #though he doesn't think it's good.
 - Henk positively evaluates this brand of salted glass. [subject-centric opinion]

We speculate that the obligatory autocentricity of *find* as well as *like* can be attributed to the interaction of experiential and causal components in their semantics. Both *find* and *like* encode an attitudinal state caused by experience. In both cases, the causal structure does not admit certain types of reasoning, such as the ban on stereotypical reasoning with *find* (48). But if that is the case, the content of the resulting state (finding something tasty or liking it) cannot be attributed to a third party, which in turn predicts autocentricity without wading into the debates on subjective language.⁴²

Thus, it is important to have a formal link between experience and opinion. Many accounts lack such a link, and therefore predict that *find* will be felicitous in cases when the finder's opinion is not based on their own experience; this is true both for the intensional accounts in Bylinina (2017) and Pearson (2013a) and the extensional account in Sæbø (2009). At the same time, any experiential account of *find* correctly predicts the pattern in (79c), as the finder is expected to ground their opinion in their experience with the object in question. Modulo the difference in implementation, we believe that the accounts in Khoo (2026) and Vardomskeya (2018) all belong to this group as they share the core intuition that *find* lexically encodes the

42. It is perhaps telling that *consider* lacks both *find*'s experiential requirement and its obligatory autocentricity.

finder’s experience.⁴³ The accounts in [Korotkova & Anand \(2021\)](#) and [Stephenson \(2007b\)](#) predict that *find* would behave like *notice* or *observe*: they require the subject to have sensory evidence for the complement proposition. This is too weak for *find*: while (79b) is felicitous when the subject observes the dragon happily consuming the food, (79c) is not. The account in [Kennedy & Willer \(2022\)](#) lies somewhere in the middle. In relativizing radical counterstances to different potential non-stipulative factors, their proposal seems to tacitly appeal to ways in which experience can ground judgment, though the characterization seems minimal enough to, for example, admit (81).

- (81) #Jake Sisko finds a career in Starfleet challenging based on ample experience of his loved ones, even though he never served himself.

Summing up, we speculate that causality may explain a rarely discussed pattern on the interpretation of subjective expressions under *find*.

5.2. Vicarious experience and vividness

In §3.5, we have argued that *find* encodes experience and that it is incompatible with certain types of grounding. One potential argument against this position comes from examples like (82): bad out of the blue (Context 1), they are felicitous in cases of narrative/imaginative transport (Context 2); see [Green & Brock \(2003\)](#) for discussion of this phenomenon at large.

- (82) The Lived Experience case

The Black Death is a 14th century epidemic of bubonic plague in Europe.

Context 1: *An out-of-the-blue remark by authors of this paper with lifespans of ordinary humans.*

Context 2: *A statement by someone who was immersed in archival work, reading historical accounts, practically living in the events, etc.*

I found the Black Death awful/devastating/disastrous.

Context 1: #, Context 2: ✓

We believe that cases like (82) do not present a counterexample to the idea that *find* requires experience, though they do force us to think more capaciously about what constitutes the relevant sort of experience. Let us suppose we give up the idea that *find* is experiential in some way. Instead, with [Kennedy & Willer \(2022\)](#), we treat *find* as imposing grounding constraints coupled with independent requirements coming from the predicate underneath (such as the acquaintance inference of *delicious*; 45b). As we have argued throughout the paper, this view does not predict that (i) *find* is bad in stereotypical reasoning cases (48); and that (ii) *find* exhibits a complex interaction with the simple present, when it can have a stative interpretation when the experience is repeatable (§4.3). We therefore maintain that *find* is experiential.

Instead, we propose that examples like (82) are best subsumed under the umbrella of vividness. Semantic and philosophical literature uses the notion of vividness to explain the behavior of predicates like *imagine* and *remember* ([Higginbotham 2003](#); [Liefke 2023, 2024](#); [Stephenson 2010](#); [Ninan 2008](#)). The upshot is as follows: when those predicates are combined with gerunds (*imagine climbing that tree*, *remember partying until dawn*), the attitude holder is required to have more than mere propositional knowledge of the prejacent proposition. Instead, it has often been claimed that the attitude holder must be an active participant of the situation

43. Though, as we note in §3.5, [Khoo \(2026\)](#) decouples strong reaction and experiential evidence.

described by the complement in case of *remember*⁴⁴ or actively picture being a participant in case of *imagine*. Significantly, *remember* can also be used in cases that Vendler (1979) calls ‘vicarious experience’: one feels so involved in the events they were not a part of that the use of *remember* is licensed despite the lack of firsthand experience. Furthermore, psychological literature suggests that vicarious experience can be processed by the brain in a way similar to actual, lived experience (Speer et al. 2009; Zwaan et al. 2002; Zwaan & Pecher 2012). When we read accounts of events, especially when strong, emotionally-charged language is used ($\approx$ thick terms), we mentally simulate the experiences of those events, which in turn may license the use of language that normally requires sensory grounding.

We propose that the behavior of *find* bears family resemblance to those cases: the finder is either a participant of the encounter event or simulates participation, which in turn explains why *find* is bad in purely non-experiential cases, such as (48). And while we do not have an explanation for vividness effects, we believe they are facilitated in situations where the finder, due to either literary technique or personal interest, is encouraged to engage in simulated experience with an object.

This is another parallel with psychological predicates. Khoo (2026:12, fn.13) argues that emotional reactions can be caused even in the absence of firsthand experience:

- (83) *You’ve heard accounts of Harvey Weinstein’s behavior towards young women in Hollywood, even though you never experienced it firsthand.*

I ??dislike/✓abhor/✓despise/✓detest Harvey Weinstein’s behavior.

(modelled after Khoo’s (2026) discussion)

Note, however, these cases too involve simulated experience, not merely the absence of experience, and are facilitated by emotionally charged expressions like *abhor*, *despise* or *detest*, in contrast to their weaker counterpart *dislike* (Khoo’s own example features *be disgusted by*, another strong term).

Summing up, we hypothesize that vividness plays a crucial role in the proper understanding of *find* as well as of psychological predicates, and points to a way to understand experience beyond the limits of our conventional descriptions. Deeper investigation on what constitutes experience is surely necessary, and will help to elucidate how experience, causality, and judgment interact in ways that admit simulated experience but ban cases of inferential reasoning.

6 Outlook

In this paper, our primary goal was to shift the focus of inquiry away from a narrow comparison between *find* and *think* on the basis of the subjectivity of their complements and to offer a comprehensive picture of the linguistic properties of *find* that, in turn, serve as a window on its semantics. We have shown that *find* shares traits of several verb classes:

- A: like other characterizing verbs (*consider*, *deem*, *judge*, *regard*), it takes small clause complements that exhibit subject-predicate asymmetries (Sæbø’s (2009) Puzzle);
- B: like emotive factives (*be amused*, *glad*, *mad*, *sad*), it blocks a true ‘de dicto’ reading in the embedded subject position and gives rise to inferences about preadjacent truth and the attitude holder’s belief;

44. Recent work argues that the evidential requirements of memory predicates are best formulated in terms of rich situations rather experience proper (Rosina 2025; Rosina & Liefke 2025a,b).

- C: *find* is a change-of-state achievement predicate, which suggests a complex event structure tracking information state update;
- D: like psychological predicates (*admire*, *abhor*, *detest*, *love*), it gives rise to an acquaintance inference rooted in the underlying event structure.

We have proposed a semantics that derives this constellation of properties in a principled way and here would like to ponder what possible meaning combinations may be lexicalized more broadly. We have argued that *find* belongs to the class of characterizing verbs and that some of its behavior can be explained by class membership, for example, the fact that it takes small clauses and gives rise to Sæbø's Puzzle. However, many properties outlined above are specific to *find*: other characterizing verbs are not factive, many of them are states, and none give rise to the same strict acquaintance inference (though many of them may have familiarity effects of the type discussed by Kennedy & Willer 2022 for *consider*). While there is vast syntactic literature on small clauses, we are not aware of any existing semantic work on characterizing predicates that attempts to delineate the nature of the class. One possibility we mentioned in §3.1 is related to information-structural constraints: if characterizing small clauses have a topic-comment structure, their subjects must be suitable topics and their predicates must be suitable comments. We leave exploring this option for future research,

On a related note, it is worth asking how to identify *find*-verbs across languages. Current work assumes that subjectivity is the defining property of what is often referred to as 'subjective attitudes'. We have demonstrated that English *find* in fact has a range of peculiar properties which may or may not be shared by its counterparts in other languages. Thus, preliminary research shows that many *find*-verbs have an acquaintance requirement (see Ducrot 1975 on French *trouver* plus small clauses, and Korotkova & Anand 2021 on several European languages). On the other hand, it is well-known that *find*-verbs have different complementation patterns: English subjective *find* requires small clauses (cf. footnote 1), while German *finden* allows adjectival clauses along with *that*-clauses (Frühauß 2015; Reis 2013) and Norwegian *synes* only takes full clauses (Sæbø 2009). One may wonder to what extent those differences in surface syntax reflect an underlying difference in meaning. For example, the complements of *synes* also exhibit subject-predicate asymmetries (Sæbø 2009) that we attributed to structured meaning, so it might be the case that *synes*-complements are closer to English small clauses than to *that*-clauses used with the Norwegian equivalent of *think*. Answering this and other questions about the cross-linguistic behavior of *find*-verbs beyond subjectivity will require careful fieldwork that pays attention to multiple facets of verbal semantics and syntax in a given language.

Finally, we have argued that *find* is a knowledge verb, not a belief verb. There is currently a renewed interest in the relationship between knowledge and belief, often analyzed via the behavior of epistemic and doxastic vocabulary (see references and discussion in Dudley & Vogel 2023). Knowledge ascriptions are known to differ from belief ascriptions along several dimensions, and only the former have been argued to be acquaintance-based (Djäv 2023; Kratzer 2002; Özyıldız 2017). As Djäv shows, this underlying semantic difference can explain DP complementation patterns: (i) *discover* is a (semi-)factive predicate, and *discover the rumor* can mean simple familiarity with the content of a rumor without endorsing it, while (ii) *believe* is a doxastic predicate, and *believe the rumor* necessarily conveys a mental attitude towards the content. The fact that *find*, unlike *think* or *believe*, gives rise to an acquaintance inference puts it squarely into the knowledge group. And while we have not discussed other uses of *find* in this paper, we want to point out a striking parallel that confirms Djäv's observation. *Find* + DP is commonly discussed in the context of physical objects, as in *find the keys* (Malink 2008;

Martin 2011; Piñón 1997; Piñon 2008). However, *find* also takes proposition-denoting nouns, as the following naturalistically occurring example demonstrates:

(84) I found that statement in a recently published textbook of psychiatry.

(*EBMH* August 2009 Vol 12 No 3)

(84) simply conveys that an entity with propositional content was located, not that the finder believes the content. This behavior is fully expected of knowledge verbs and aligns with the idea that acquaintance knowledge and propositional knowledge can be treated uniformly; it is difficult to explain if *find* is a belief verb. We speculate that *find* encodes an event of encounter across its distinct uses and hope to see more research on verbal polysemy in the future. This type of work, uncommon in the research on the language of mental attitudes, can help elucidate the concepts expressed by attitude predicates, which in turn addresses the fundamental questions that underpins most work on *find*: what kind of beast it is and what it tells us about the nature of attitude ascriptions.

References

- Abrusán, Márta. 2011. Presuppositional and negative islands: a semantic account. *Natural Language Semantics* 19. 257–321.
- Abusch, Dorit. 1997. Sequence of tense and temporal *de re*. *Linguistics and Philosophy* 20. 1–50.
- Anand, Pranav & Valentine Hacquard. 2014. Factivity, belief and discourse. In Luka Crnić & Uli Sauerland (eds.), *The Art and Craft of Semantics: A Festschrift for Irene Heim*, 69–90. Cambridge, MA: MITWPL.
- Anand, Pranav & Natasha Korotkova. 2018. Acquaintance content and obviation. In Uli Sauerland & Stephanie Solt (eds.), *Sinn und bedeutung* 22, 161–173. Berlin: ZAS. doi:10.18148/sub/2018.v22i1.65.
- Anand, Pranav & Natasha Korotkova. 2022. How to theorize about subjective meaning: A lesson from ‘*de re*’. *Linguistics and Philosophy* 45(3). 619–681. doi:10.1007/s10988-021-09331-0.
- Anand, Pranav & Natasha Korotkova. 2024. Facts, intentions, questions: English “come-to-know” predicates in deliberative environments. In Fausto Carcassi, Tamar Johnson, Soren Brinck Knudstorp, Sabina Domínguez Parrado, Pablo Rivas Robledo & Giorgio Sbardolini (eds.), *Proceedings of the 24th amsterdam colloquium*, 15–21. Amsterdam: ILLC.
- Asher, Nicholas. 2011. *Lexical meaning in context: A web of words*, vol. 125 Cambridge Studies in Linguistics. Cambridge: Cambridge University Press.
- Asher, Nicholas. 2014. Selectional restrictions, types and categories. *Journal of Applied Logic* 12(1). 75–87. doi:10.1016/j.jal.2013.08.002.
- Bade, Nadine. 2021. On the scope and nature of *Maximise Presupposition*. *Language and Linguistics Compass* 15(6). doi:10.1111/lnc3.12416.
- Baglini, Rebekah & Elitzur A. Bar-Asher Siegal. 2025. Modeling linguistic causation. *Linguistics and Philosophy* 48. 647–691. doi:10.1007/s10988-025-09436-w. <https://doi.org/10.1007/s10988-025-09436-w>.
- Bar-Asher Siegal, Elitzur A. 2026. Causation and causal relations: Insights from natural language. *Annual Review of Linguistics* 12(1). 125–145. doi:10.1146/annurev-linguistics-041824-031335.
- Barker, Chris. 2013. Negotiating taste. *Inquiry: An Interdisciplinary Journal of Philosophy* 56(2-3). 240–257.
- Bartsch, Renate. 1973. “negative transportation” gibt es nicht. *Linguistische Berichte* 27. 1–7.
- Barwise, Jon. 1981. Scenes and other situations. *Journal of Philosophy* 78(7). 369–397.
- Basilico, David. 2003. The topic of small clauses. *Linguistic Inquiry* 34(1). 1–35.
- Beavers, John & Andrew Koontz-Garboden. 2020. *The roots of verbal meaning*. Oxford: Oxford University Press. doi:10.1093/oso/9780198855781.001.0001. <https://doi.org/10.1093/oso/9780198855781.001.0001>.
- Bittner, Maria. 1998a. Concealed causatives. *Natural Language Semantics* 7(1). 1–78. doi:10.1023/a:1008313608525.
- Bittner, Maria. 1998b. Cross-linguistic semantics for questions. *Linguistics and Philosophy* 21(1). 1–82. doi:10.1023/a:1005313305414.
- Bogal-Albritten, Elizabeth. 2016. *Building meaning in Navajo*: University of Massachusetts, Amherst dissertation.
- Bondarenko, Tatiana, Richard Luo & Vincent Rouillard. 2024. Thinking statively and dynamically: A view from Georgian. In Fausto Carcassi, Tamar Johnson, Soren Brinck Knudstorp, Sabina Domínguez Parrado, Pablo Rivas Robledo & Giorgio Sbardolini (eds.), *Proceedings of the 24th amsterdam colloquium*, 44–50. Amsterdam: ILLC.
- Boneh, Nora. 2019. Dispositions and characterizing sentences. *Glossa: A Journal of General Linguistics* 4(1). doi:10.5334/gjgl.540.

- Borkin, Ann. 1973. To be or not to be. In *The 9th regional meeting of the chicago linguistics society*, 44–56. Chicago, IL.
- Bouchard, David-Etienne. 2012. *Long-distance degree quantification and the grammar of subjectivity*: McGill University dissertation.
- Briesen, Jochen. 2024. Aesthetic judgments, evaluative content, and (hybrid) expressivism. *Ergo an Open Access Journal of Philosophy* 11(19). doi:10.3998/ergo.6159.
- Bylinina, Lisa. 2017. Judge-dependence in degree constructions. *Journal of Semantics* 34(2). 291–331. doi:10.1093/jos/ffw011.
- Cable, Seth. 2022. Two paths to habituality: The semantics of habitual mode in tlingit. *Semantics & Pragmatics* 15(11). 1–55. doi:10.3765/sp.15.11. <https://doi.org/10.3765/sp.15.11>.
- Cardinaletti, Anna & Maria Teresa Guasti. 1995. Small clauses: Some controversies and issues of acquisition. In Anna Cardinaletti & Maria Teresa Guasti (eds.), *Syntax and semantics* 28, 1–27. San Diego, CA: Academic Press.
- Cariani, Fabrizio. 2022. Future-past asymmetries, evidential grounding, and projection. In Marco Degano, Tom Roberts, Giorgio Sbardolini & Marieke Schouwstra (eds.), *Proceedings of the 23rd amsterdam colloquium*, 62–68. ILLC.
- Carlson, Gregory N. 1980. *Reference to kinds in english*. New York: Garland.
- Choi, Sungho & Michael Fara. 2021. Dispositions. In Edward N. Zalta (ed.), *The Stanford encyclopedia of philosophy*, Metaphysics Research Lab, Stanford University Spring 2021 edn.
- Cohen, Ariel & Nomi Erteschik-Shir. 2002. Topic, focus, and the interpretation of bare plurals. *Natural Language Semantics* 10(2). 125–165.
- Collins, John. 2013. The syntax of personal taste. *Philosophical perspectives* 27(1). 55–103.
- Coppock, Elizabeth. 2018. Outlook-based semantics. *Linguistics and Philosophy* 41(2). 125–164.
- Crespo, Inés & Frank Veltman. 2019. Tasting and testing. *Linguistics and Philosophy*.
- Djävrv, Kajsa. 2019. *Factive and assertive attitude reports*: University of Pennsylvania dissertation. <https://repository.upenn.edu/edissertations/3645>.
- Djävrv, Kajsa. 2023. Knowing and believing things: What DP-complements can tell us about the argument structure and composition of (factive) attitudes. *Journal of Semantics* 40(2). 1–55. doi:doi.org/10.1093/jos/ffac015.
- Dowty, David R. 1979. *Word meaning and montague grammar*. Dordrecht: D. Reidel.
- Ducrot, Oswald. 1975. Je trouve que. *Semantikos* 1(1). 63–88.
- Dudley, Rachel & Christopher Vogel. 2023. Knowledge, the concept KNOW, and the word know: considerations from polysemy and pragmatics. *Synthese* 203(1). doi:10.1007/s11229-023-04439-1.
- Egan, Andy. 2010. Disputing about taste. In R. Feldman & T. A. Warfield (eds.), *Disagreement*, 247–286. Oxford University Press. doi:<https://doi.org/10.1093/acprof:oso/9780199226078.003.0011>.
- Egan, Andy, John Hawthorne & Brian Weatherson. 2005. Epistemic modals in context. In Gerhard Preyer & Georg Peter (eds.), *Contextualism in philosophy: Knowledge, meaning and truth*, 131–169. Oxford: Oxford University Press.
- Erteschik-Shir, Nomi. Forth. Information structure and (in)definiteness. In Patricia Cabredo Hofherr & Anne Zribi-Hertz (eds.), *Languages with and without articles*, Emerald.
- Farkas, Donka. 1997. Evaluation indices and scope. In Anna Szabolcsi (ed.), *Ways of scope taking*, 183–215. Dordrecht: Springer Netherlands.
- Ferreira, Marcelo. 2016. The semantic ingredients of imperfectivity in progressives, habituals, and counterfactuals. *Natural Language Semantics* 24(4). 353–397. doi:10.1007/s11050-016-9127-2. <https://doi.org/10.1007/s11050-016-9127-2>.
- Filip, Hana. 2020. Lexical aspect (aktionsart). In Lisa Matthewson, Cécile Meier, Hotze Rullman & Thomas Ede Zimmerman (eds.), *The blackwell companion to semantics*, 1–61. Wiley. doi:10.1002/9781118788516.sem038.

- Fodor, Janet Dean. 1970a. *The linguistic description of opaque contexts*: Massachusetts Insitute of Technology dissertation.
- Fodor, Jerry A. 1970b. Three reasons for not deriving ‘kill’ from ‘cause to die’. *Linguistic Inquiry* 1. 429–438.
- Franzén, Nils. 2018. Aesthetic evaluation and first-hand experience. *Australasian Journal of Philosophy* 96(4). 669–682.
- Franzén, Nils. 2020. Evaluative discourse and affective states of mind. *Mind* 129(516). 1095–1126.
- Franzén, Nils. 2021. Non-factualism and evaluative supervenience. *Inquiry* 1–22.
- Frühauf, Felix. 2015. *Bedeutung und Verwendung des deutschen Einschätzungsverbs finden und seine Entsprechung im Englischen*: Humboldt University, Berlin BA thesis.
- Fusco, Achille. 2025. Responsible attitudes: The case of Italian ‘considerare’. In Federica Longo & Daniele Panizza (eds.), *Proceedings of sinn und bedeutung 29*, 534–551. doi:10.18148/SUB/2025.V29.1229.
- Fusco, Achille, Cristiano Chesi & Valentina Bianchi. 2025. Using belief-perception mismatch to assess the meaning of subjective attitude verbs. In Gabriela Bîlbîe & Gerhard Schaden (eds.), *Empirical issues in syntax and semantics: Selected papers from cssp 2023*, 91–112. Berlin: Language Science Press. doi:10.5281/ZENODO.15450434.
- Gajewski, Jon. 2007. Neg-raising and polarity. *Linguistics and Philosophy* 30. 289–328.
- Gibbard, Allan. 1990. *Wise choices, apt feelings*. Oxford: Clarendon Press.
- Glanzberg, Michael. 2007. Context, content, and relativism. *Philosophical Studies* 136(1). 1–29.
- Green, Melanie C. & Timothy C. Brock. 2003. In the mind’s eye: Transportation-imagery model of narrative persuasion. In Melanie C. Green, Jeffrey J. Strange & Timothy C. Brock (eds.), *Narrative impact: Social and cognitive foundations*, 337–364. Psychology Press. doi:10.4324/9781410606648-22.
- Grimshaw, Jane. 1979. Complement selection and the lexicon. *Linguistic Inquiry* 10(2). 279–326.
- Grob, Dana. 2022. *Ambiguity with Almost and the scalar structure of accomplishments*: Hebrew University of Jerusalem MA thesis.
- Gyarmathy, Zsófia. 2015. *Achievements, durativity and scales*: University of Düsseldorf dissertation.
- Hacquard, Valentine. 2010. On the event relativity of modal auxiliaries. *Natural Language Semantics* 18(1). 79–114.
- Heim, Irene. 1991. Artikel und Definitheit [Articles and definiteness]. In Arnim von Stehow & Dieter Wunderlich (eds.), *Semantik: Ein internationales Handbuch derzeitgenössischen Forschung*, 487–535. Berlin: de Gruyter.
- Heyde-Zybatov, Tatjana. 2008. What kind of events do achievements describe? In Anita Staube (ed.), *The discourse potential of underspecified structures*, 109–144. Walter de Gruyter.
- Higginbotham, James. 1983. The logic of perceptual reports: An extensional alternative to situation semantics. *The Journal of Philosophy* 80(2). 100–127.
- Higginbotham, James. 2003. Remembering, imagining, and the first person. In A. Barber (ed.), *Epistemology of language*, 496–533. Oxford University Press.
- Jäger, Gerhard. 2001. Topic-comment structure and the contrast between stage level and individual level predicates. *Journal of Semantics* 18(2). 83–126. doi:10.1093/jos/18.2.83.
- Karttunen, Lauri. 1971. Some observations on factivity. *Papers in Linguistics* 4(1). 55–69.
- Karttunen, Lauri. 1973. Presuppositions of compound sentences. *Linguistic inquiry* 4(2). 169–193.
- Katz, Jerrold J. & Jerry A. Fodor. 1963. The structure of a semantic theory. *Language* 39(2). 170–210.
- Kearns, Kate. 2003. Durative achievements and individual-level predicates on events. *Linguistics and Philosophy* 26. 595–635.

- Kennedy, Christopher. 2013. Two sources of subjectivity: Qualitative assessment and dimensional uncertainty. *Inquiry* 56(2-3). 258–277.
- Kennedy, Christopher & Malte Willer. 2016. Subjective attitudes and counterstance contingency. In Mary Moroney, Carol-Rose Little, Jacob Collard & Dan Burgdorf (eds.), *Semantics and linguistic theory* 26, 913–933. doi:10.3765/salt.v26i0.3936.
- Kennedy, Christopher & Malte Willer. 2022. Familiarity inferences, subjective attitudes and counterstance contingency: Towards a pragmatic theory of subjective meaning. *Linguistics and Philosophy* 45. 1395–1445. doi:10.1007/s10988-022-09358-x.
- Khoo, Justin. 2025. Judging for ourselves. *Philosophy and Phenomenological Research* 110(3). 757–788. doi:10.1111/phpr.13133.
- Khoo, Justin. 2026. Finding and believing. *Meaning: A Journal of Linguistics and Philosophy* 2. 1–24. doi:10.46586/meaning.2026.a1.
- Kiparsky, Paul & Carol Kiparsky. 1970. Fact. In M. Bierwisch & K. Heidolph (eds.), *Progress in linguistics*, 143–173. Mouton de Gruyter.
- Kölbel, Max. 2004. Faultless disagreement. *Proceedings of the Aristotelian Society* 104. 53–73.
- Korotkova, Natasha. Forth. The notional category of evidentiality. In Ana Arregui, Valentine Hacquard & Michela Ippolito (eds.), *The cambridge handbook of natural language modality*, Cambridge University Press.
- Korotkova, Natasha & Pranav Anand. 2021. Find, must and conflicting evidence. In Patrick Georg Grosz, Luisa Martí, Hazel Pearson, Yasutada Sudo & Sarah Zobel (eds.), *Proceedings of sinn und bedeutung* 25, 515–532. doi:10.18148/sub/2021.v25i0.951.
- Korotkova, Natasha & Pranav Anand. 2024. Today we learned: English ‘come-to-know’ attitudes as a natural class. Ms., Utrecht University and UC Santa Cruz.
- Kratzer, Angelika. 2002. Facts: Particulars or information units? *Linguistics and Philosophy* 25. 655–670. doi:doi:10.1023/A:1020807615085.
- Krawczyk, Elizabeth A. 2012. *Inferred propositions and the expression of the evidence relation in natural language. evidentiality in central alaskan yup’ik eskimo and english*: Georgetown University dissertation.
- Krifka, Manfred. 1992. A compositional semantics for multiple focus constructions. In Joachim Jacobs (ed.), *Informationsstruktur und grammatik*, 17–53. Wiesbaden: VS Verlag für Sozialwissenschaften. doi:10.1007/978-3-663-12176-3_2.
- Krifka, Manfred. 2001. For a structured meaning account of questions and answers. In Caroline Féry & Wolfgang Sternefeld (eds.), *Audiatur vox sapientiae: A festschrift for Arnim von Stechow*, 287–319. Berlin, Boston: Akademie Verlag. doi:10.1515/9783050080116.287.
- Krifka, Manfred, Francis Jeffry Pelletier, Gregory N. Carlson, Alice ter Meulen, Gennaro Chierchia & Godehard Link. 1995. Genericity: An introduction. In Gregory N. Carlson & Francis Jeffry Pelletier (eds.), *The generic book*, 1–124. Chicago: University of Chicago Press.
- Kuroda, Sige-Yuki. 1972. The categorical and the thetic judgment: Evidence from Japanese syntax. *Foundations of Language* 9(2). 153–185.
- Kuroda, Sige-Yuki. 1990. The categorical and the thetic judgment reconsidered. In K. Mulligan (ed.), *Mind, meaning and metaphysics*, 77–88. Dordrecht: Springer Netherlands. doi:10.1007/978-94-009-0505-4_6.
- Ladusaw, William A. 1994. Thetic and categorical, stage and individual, weak and strong. In Mandy Harvey & Lynn Santelmann (eds.), *Semantics and linguistic theory* 4, 220–229. Linguistic Society of America.
- Lande, Inna. 2009. *Subjektive Einstellungsberichte im Deutschen: Einstellungsverben und beurteilungsabhängige Prädikate*: University of Oslo MA thesis.

- Lasersohn, Peter. 2005. Context dependence, disagreement, and predicates of personal taste. *Linguistics and Philosophy* 28(6). 643–686. doi:10.1007/s10988-005-0596-x.
- Liefke, Kristina. 2023. Experiential attitude reports. *Philosophy Compass* 18(6). 1–17. doi:10.1111/phc3.12913.
- Liefke, Kristina. 2024. Just simulating? Linguistic support for continuism about remembering and imagining. *Review of Philosophy and Psychology* 16(3). 745–781. doi:10.1007/s13164-024-00751-y.
- Lohninger, Magdalena, Kovač Iva & Susanne Wurmbrand. 2022. From prolepsis to hyperraising. *Philosophies* 7(32). 1–40. doi:10.3390/philosophies7020032.
- MacFarlane, John. 2014. *Assessment Sensitivity: Relative Truth and Its Applications*. Oxford: Oxford University Press.
- Malink, Marko. 2008. Right boundary achievements under conative negation. In Anita Steube (ed.), *The discourse potential of underspecified structures*, 145–162. Berlin, New York: De Gruyter. doi:10.1515/9783110209303.2.145. <https://doi.org/10.1515/9783110209303.2.145>.
- Martin, Fabienne. 2011. Revisiting the distinction between accomplishments and achievements. In Tanja Mortelmans, Jesse Mortelmans & Walder De Mulder (eds.), *From now to eternity*, 43–64. Leiden: Brill. doi:10.1163/9789042032682_004.
- McNally, Louise. 2020. Strong and weak nominals. In Lisa Matthewson, Cécile Meier, Hotze Rullman & Thomas Ede Zimmerman (eds.), *The blackwell companion to semantics*, 1–26. Wiley. doi:10.1002/9781118788516.sem084.
- McNally, Louise. 2024. Two approaches to the semantic decomposition of change of state verbs, how they differ, and why it matters. *Glossa: A Journal of General Linguistics* 9(1). doi:10.16995/glossa.9948. <https://doi.org/10.16995/glossa.9948>.
- Mendes, Jéssica & Caleb Kendrick. 2024. Composing *find*. Talk at the workshop “(De-)composition and modification of attitude predicates”, DGfS 46.
- Menéndez-Benito, Paula. 2013. On dispositional sentences. In Alda Mari, Claire Beyssade & Fabio del Prete (eds.), *Genericity*, 276–292. Oxford: Oxford University Press.
- Moulton, Keir. 2009. *Natural selection and the syntax of clausal complementation*: UMass, Amherst dissertation.
- Muñoz, Patrick. 2019. *On tongues: The grammar of experiential evaluation*: University of Chicago dissertation.
- Musan, Renate. 1997a. *On the Temporal Interpretation of Noun Phrases*. New York: Garland.
- Musan, Renate. 1997b. Tense, predicates, and lifetime effects. *Natural Language Semantics* 5(3). 271–301. doi:10.1023/a:1008281017969.
- Nadathur, Prerna. 2023. *Actuality inferences: Causality, aspect, and modality*. Oxford: Oxford University Press.
- Nadathur, Prerna & Bar-Asher Siegal. 2025. Modeling progress: causal models and the imperfective paradox. Ms., <https://ling.auf.net/lingbuzz/009209>.
- Neeleman, Ad & Hans van de Koot. 2012. The linguistic expression of causation. In Martin Everaert, Marijana Marelj & Tal Siloni (eds.), *The theta system: Argument structure at the interface*, 20–51. Oxford: Oxford University Press.
- Ninan, Dilip. 2008. *Imagination, content, and the self*: Massachusetts Institute of Technology dissertation.
- Ninan, Dilip. 2014. Taste predicates and the acquaintance inference. In Todd Snider, Sarah D’Antonio & Mia Weigand (eds.), *Semantics and linguistic theory* 24, 290–309. LS.
- Ninan, Dilip. 2020. The projection problem for predicates of personal taste. In Joseph Rhyne, Kaelyn Lamp, Nicole Dreier & Chloe Kwon (eds.), *Semantics and linguistic theory* 30, 753–778. LSA.
- Ninan, Dilip. 2024. An expressivist theory of taste predicates. *Philosophers’ Imprint* 24(1). doi:10.3998/phimp.2683.

- Ogihara, Toshiyuki & Yael Sharvit. 2012. Embedded tenses. In Robert I. Binnick (ed.), *The Oxford handbook of tense and aspect*, chap. 22, 638–668. OUP.
- Özyıldız, Deniz. 2017. Attitude reports with and without true belief. In Dan Burgdorf, Jacob Collard, Sireemas Maspong & Brynhildur Stefánsdóttir (eds.), *Proceedings of SALT 27*, 397–417.
- Özyıldız, Deniz. 2021. *The event structure of attitudes*: University of Massachusetts, Amherst dissertation.
- Pearson, Hazel. 2013a. A judge-free semantics for predicates of personal taste. *Journal of Semantics* 30(1). 103–154. doi:<https://doi.org/10.1093/jos/ffs001>.
- Pearson, Hazel. 2013b. *The sense of self: topics in the semantics of de se expressions*: Harvard dissertation.
- Pearson, Hazel. 2022. Individual- and stage-level predicates of personal taste. In Jeremy Wyatt, Julia Zakkou & Dan Zeman (eds.), *Perspectives on taste*, Routledge.
- Percus, Orin. 2000. Constraints on some other variables in syntax. *Natural Language Semantics* 8(3). 173–229.
- Piñón, Christopher. 1997. Achievements in event semantics. In Aaron Lawson (ed.), *Semantics and linguistic theory* 7, 276–293. Ithaca, NY: Cornell university.
- Piñón, Christopher. 2008. Negating right boundary achievements. In Anita Staube (ed.), *The discourse potential of underspecified structures*, 163–175. Walter de Gruyter.
- Plunkett, David & Timothy Sundell. 2013. Disagreement and the semantics of normative and evaluative terms. *Philosophers' Imprint* 13.
- Portner, Paul. 1998. The progressive in modal semantics. *Language* 74(4). 760–787. doi:10.1353/lan.1998.0219. <https://doi.org/10.1353/lan.1998.0219>.
- Pustejovsky, James. 1995. *The generative lexicon*. Cambridge, MA: MIT Press.
- Quine, Willard Van Orman. 1956. Quantifiers and propositional attitudes. *Journal of Philosophy* 53. 101–111.
- Raposo, Eduardo & Juan Uriagereka. 1995. Two types of small clauses. In Anna Cardinaletti & Maria Teresa Guasti (eds.), *Syntax and semantics 28: Small clauses*, 179–206. San Diego, CA: Academic Press.
- Rawlins, Kyle & Aaron White. 2019. Question agnosticism and change of state. In Robert Truswell, Chris Cummins, Caroline Heycock, Brian Rabern & Hannah Rohde (eds.), *Proceedings of sinn und bedeutung* 21, 1325–1342.
- Reis, Marga. 2013. Dt. finden und “subjektive Bedeutung”. *Linguistische Berichte* 2013(236). 389–426.
- Rosina, Emil & Kristina Liefke. 2025a. German ‘noch genau wissen’: Uniform semantics, distinct effects. In Shweta Akolkar, Amber Galvano, Akil Ismael, Kang Franco Liu & Line Mikkelsen (eds.), *Proceedings of the 42nd west coast conference on formal linguistics*, 330–337. Somerville, MA: Cascadilla Press.
- Rosina, Emil Eva. 2025. On using experiments on remember as evidence in formal semantics. *Linguistica Brunensia* 73(2). 35–72. doi:10.5817/lb2025-38652.
- Rosina, Emil Eva & Kristina Liefke. 2025b. Experientiality markers in memory reports: A semantics-pragmatics puzzle. In *Proceedings of experiments in linguistic meaning* 3, vol. 3, 319–331. Linguistic Society of America. doi:10.3765/elm.3.5815.
- Rothstein, Susan. 2004. *Structuring events: A study in the semantics of lexical aspect*. Wiley-Blackwell.
- Schwager, Magdalena. 2009. Speaking of qualities. In Ed Cormany, Satoshi Ito, & David Lutz (eds.), *Proceedings of SALT 19*, 395–412. doi:10.3765/salt.v19i0.2534.
- Scontras, Gregory & Judith Tonhauser. 2026. Predicting within-utterance projection variation. *Meaning: A journal of linguistics and philosophy* 2. 1–27. doi:10.46586/meaning.2026.a2.
- Sharvit, Yael & Elena Guerzoni. 2003. Reconstruction and its problems. In Paul Dekker & Robert van Rooij (eds.), *Proceedings of the 14th amsterdam colloquium*, 205–210. Amsterdam: ILLC.

- Shibatani, Masayoshi. 1976. The grammar of causative constructions: A conspectus. In Masayoshi Shibatani (ed.), *The grammar of causative constructions*, 1–39. New York: Academic Press.
- Simons, Mandy, Judith Tonhauser, David Beaver & Craige Roberts. 2010. What projects and why. In Nan Li & David Lutz (eds.), *Semantics and Linguistic Theory (SALT) 20*, 309–327. Ithaca, NY: LSA and CLC Publications. doi:<https://doi.org/10.3765/salt.v20i0.2584>.
- Solt, Stephanie. 2018. Multidimensionality, subjectivity and scales: Experimental evidence. In Elena Castroviejo, Louise McNally & Galit Weidman Sassoon (eds.), *The semantics of gradability, vagueness, and scale structure: Experimental perspectives*, 59–91. Cham: Springer.
- Soria-Ruiz, Andrés. 2021. Evaluative and metalinguistic dispute. *Australasian Journal of Philosophy* 101(1). 165–181. doi:[10.1080/00048402.2021.1959624](https://doi.org/10.1080/00048402.2021.1959624).
- Soria-Ruiz, Andrés. 2024. Expressive inferences and mighty obstacles. In Fausto Carcassi, Tamar Johnson, Soren Brinck Knudstorp, Sabina Domínguez Parrado, Pablo Rivas Robledo & Giorgio Sbardolini (eds.), *Proceedings of the 24th amsterdam colloquium*, 358–364. Amsterdam: ILLC.
- Speer, Nicole K., Jeremy R. Reynolds, Khena M. Swallow & Jeffrey M. Zacks. 2009. Reading stories activates neural representations of visual and motor experiences. *Psychological Science* 20(8). 989–999. doi:[10.1111/j.1467-9280.2009.02397.x](https://doi.org/10.1111/j.1467-9280.2009.02397.x).
- von Stechow, Arnim. 1991. Focusing and background operators. In Werner Abraham (ed.), *Discourse particles: Descriptive and theoretical investigations on the logical, syntactic and pragmatic properties of discourse particles in german*, 37–84. John Benjamins Publishing Company. doi:[10.1075/pbns.12.04ste](https://doi.org/10.1075/pbns.12.04ste).
- Stephenson, Tamina. 2007a. Judge dependence, epistemic modals, and predicates of personal taste. *Linguistics and Philosophy* 30(4). 487–525. doi:[10.1007/s10988-008-9023-4](https://doi.org/10.1007/s10988-008-9023-4).
- Stephenson, Tamina. 2007b. *Towards a Theory of Subjective Meaning*: Massachusetts Institute of Technology dissertation.
- Stephenson, Tamina. 2010. Vivid attitudes: Centered situations in the semantics of “remember” and “imagine”. In Nan Li & David Lutz (eds.), *Semantics and linguistics theory 20*, 147–160. Linguistic Society of America. doi:[10.3765/salt.v20i0.2582](https://doi.org/10.3765/salt.v20i0.2582).
- Sæbø, Kjell Johan. 2009. Judgment ascriptions. *Linguistics and Philosophy* 32(4). 327–352.
- Stojanovic, Isidora. 2007. Talking about taste: Disagreement, implicit arguments, and relative truth. *Linguistics and Philosophy* 30(6). 691–706.
- Stojanovic, Isidora & Louise McNally. 2023. Are moral predicates subjective? A corpus study. In David Bordonaba-Plou (ed.), *Experimental philosophy of language: Perspectives, methods, and prospects*, 103–120. Cham: Springer.
- Tomioka, Satoshi. 2021. Strong and weak nominals. In Lisa Matthewson, Cécile Meier, Hotze Rullman & Thomas Ede Zimmerman (eds.), *Topic*, 1–26. Wiley. doi:[10.1002/9781118788516.sem125](https://doi.org/10.1002/9781118788516.sem125).
- Tonhauser, Judith. 2012. Diagnosing (not-)at-issue content. In Elizabeth Bogal-Allbritten (ed.), *Proceedings of the Sixth Conference on the Semantics of Under-Represented Languages in the Americas (SULA)*, 239–254. Amherst, MA: GLSA Publications.
- Tonhauser, Judith, David Beaver, Craige Roberts & Mandy Simons. 2013. Towards a taxonomy of projective content. *Language* 89(1). 66–109. doi:<https://doi.org/10.1353/lan.2013.0001>.
- Vardomskaya, Tamara. 2018. *Sources of Subjectivity*: University of Chicago dissertation.
- Vendler, Zeno. 1957. Verbs and times. *Philosophical Review* 56. 143–160.
- Vendler, Zeno. 1979. Vicarious experience. *Revue de Métaphysique et de Morale* 84(2). 161–173.
- van Wijnbergen-Huitink, Janneke. 2016. Subjective meaning: An introduction. In Cécile Maier & Janneke van Wijnbergen-Huitink (eds.), *Subjective meaning: Alternatives to relativism* 1–19, Berlin/Boston: De Gruyter.
- Williams, Edwin. 1975. Small clauses in English. In J.P. Kimball (ed.), *Syntax and semantics 4*, 249–273. Orlando: Academic Press.

- Wolff, Phillip. 2003. Direct causation in the linguistic coding and individuation of causal events. *Cognition* 88. 1–48.
- Wurmbrand, Susanne & Magdalena Lohninger. 2023. An implicational universal in complementation: Theoretical insights and empirical progress. In Yutta Hartmann & Angelika Wöllstein (eds.), *Propositionale argumente im sprachvergleich: Theorie und empirie*, 183–229. Tübingen: Gunter Narr Verlag.
- Wurmbrand, Suzanne. 2024. The size of clausal complements. *Annual Reviews of Linguistics* 10. 59–83. doi:10.1146/annurev-linguistics-031522-103802.
- Yoon, James H. 2007. Raising of major arguments in Korean and Japanese. *Natural Language and Linguistic Theory* 25(3). 615–653. doi:10.1007/s11049-007-9020-2.
- Zakkou, Julia. 2019. *Faultless disagreement: A defense of contextualism in the realm of personal taste*. Frankfurt am Main: Vittorio Klostermann.
- Zuchewicz, Karolina. 2020. *On the veridicality of perfective clause-embedding verbs in Polish: A unified aspect-based analysis of incremental theme verbs with nominal and propositional complements*. Humboldt University, Berlin dissertation.
- Zwaan, Rolf A. & Diane Pecher. 2012. Revisiting mental simulation in language comprehension: Six replication attempts. *PLoS ONE* 7(12). e51382. doi:10.1371/journal.pone.0051382.
- Zwaan, Rolf A., Robert A. Stanfield & Richard H. Yaxley. 2002. Language comprehenders mentally represent the shapes of objects. *Psychological Science* 13(2). 168–171.